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Light vectors and scalar interactions near heterotic K3 gauge enhancement

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Abstract: We prove a uniform comparison for a four-scalar tree amplitude near gauge enhancement in a smooth $SO(32)$ heterotic K3 compactification. A rank-six stable-bundle family approaches the direct sum of two isomorphic stable rank-three bundles. Two harmonic deformation amplitudes a, b give three squared Higgs masses proportional to b^2, a^2 and $a^2 + b^2$. For two specified massless bundle scalars, the complete leading Yang–Mills amplitude, including all internal vector and scalar Kaluza–Klein modes, approaches an explicit light-vector exchange formula. The absolute error is quadratic in the deformation amplitudes, uniformly as b/a tends to zero, below the heavier Higgs masses and away from the light poles. Harmonic orthogonality controls the actual cubic vertices, while an exact scalar contact–exchange cancellation removes inverse-mass losses. The same estimates give inverse-square growth of the sectional curvature of the harmonic bundle-moduli metric on a selected plane, with uniform relative error. We also obtain a ratio-uniform relative bound for the first-order coupled vector equation and a fixed-ray coefficient with a Pauli selection rule. At the next tree order we control quadratic potential mixing with K3 metric deformations and compute the universal graviton exchange. The complete coefficient at that order and a finite-string-scale remainder remain open.

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Contents

1 Introduction **1**

2 The compactification and its enhancement locus **4**
2.1 Stable factors and the orthogonal embedding 4
2.2 Harmonic coordinates and actual bundles 5

3 Light vectors and the retained spectrum **7**
3.1 The physical vector operator 7
3.2 Massless fields and Higgs multiplets 8
3.3 The common low-energy space 9

4 The first coupled vector correction **10**
4.1 Action convention and the corrected background 10
4.2 The two heavy spin-one sectors 11
4.3 Hodge cancellation, positivity and the error bound 12
4.4 A relative estimate uniform in the Higgs hierarchy 13

5 Actual scalar modes and their gauge vertices **15**
5.1 Harmonic representatives of the external states 15
5.2 The light eigenvector 16
5.3 Uniform cubic vertices 17

6 Scattering across the light-vector scale **18**
6.1 Scalar partners and the complete connection spectrum 19
6.2 The spectral amplitude and contact cancellation 19
6.3 A comparison uniform in the Higgs hierarchy 20
6.4 Uniform curvature of the selected plane 21

7 The leading physical curvature coefficient **23**
7.1 Partial reduction on a regular patch 23
7.2 Mixed constraints and the vertical kinetic coefficient 24
7.3 The other neutral fields and ambient curvature 25
7.4 The interaction and its order of limits 26

8 Partial results at the next heterotic tree order **27**
8.1 Order counting and the required terms 27
8.2 Quadratic mixing with K3 metric deformations 28
8.3 The massless graviton and the angular domain 29

9 Discussion 30

A The vector correction on a regular ray 31

B Regular-patch derivatives and the quotient model 33
B.1 Preserving the quadratic rate in holomorphic coordinates 33
B.2 What fails near a changing stabilizer 35
B.3 Curvature in additional harmonic deformation directions 35

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1 Introduction

Near gauge enhancement, a heterotic compactification can have several Higgs scales that vanish at unrelated rates. Scalar scattering then probes different effective descriptions as the energy crosses those scales. Below the smallest mass, the massive vector multiplets can be eliminated and their interactions are encoded in a quotient metric. Above that mass, the light multiplet must remain explicit. Establishing this transition on an actual compactification requires control of its normalized states, vertices and complete internal exchanges, in addition to its masses.

The enhanced vectors arise from parallel adjoint sections of the internal connection. The finite-dimensional Higgs mass matrix describes how they become massive, but an absolute approximation need not resolve the smallest eigenvalue. Similarly, a comparison of scalar kinetic metrics does not control their curvature by differentiation, or identify a bundle-moduli curvature with an interaction in the full reduced theory. We obtain relative mass estimates and a uniform scattering comparison by working directly with the reconstructed connection and its harmonic modes.

We address these questions on one family of smooth K3 compactifications of the $\mathrm{Spin}(32)/\mathbb{Z}_2$ heterotic theory with vector structure. We use $\mathrm{SO}(32)$ for its gauge algebra and its real vector-bundle description. The background bundle contains two identical stable rank-three factors at the enhancement point. Two orthonormal harmonic deformation coordinates, with coefficients a, b, break the multiplicity $\mathrm{SU}(2)$. The K3 surface remains smooth and its polarization is fixed. The moment maps vanish throughout this bundle enhancement; no nonzero Fayet–Iliopoulos parameter is varied.

Let g_0 be the fixed reference K3 metric, $V_0 = \mathrm{Vol}(g_0)$, and let $L^2 g_0$ be the physical internal metric at leading order. The three leading squared vector masses obey

$$m_j^{2,(0)} = \widehat{m}_j^2\big(1 + O(a^2 + b^2)\big), \qquad (\widehat{m}_1^2, \widehat{m}_2^2, \widehat{m}_3^2) = \frac{8}{3V_0 L^2}(b^2, a^2, a^2 + b^2), \quad 0 < b \le a. \tag{1.1}$$

The error is proportional to each individual squared mass, with a constant independent of b/a. In particular, at $a = \varepsilon$, $b = \varepsilon^p$, $p > 1$, the weakest squared mass has order ε^{2p}/L^2, while

its relative geometric error still has order ε^2. This is a statement about actual Hermitian–Einstein connections, obtained by applying the quantitative local comparison of [1] to the compactification family. The Pauli-matrix eigenvalues alone would not give this conclusion.

We also calculate the first coefficient of the coupled tree-level vector equations. Write $\eta = \alpha'/L^2$. Eliminating the complete leading metric and two-form vector towers gives a finite, energy-dependent pencil $F^{[1]}(\lambda, \eta)$, defined in section 4. Its three positive roots, converted to squared masses by L^{-2}, satisfy

$$\left| \frac{m_j^{2,[1]}}{\widehat{m}_j^2} - 1 \right| \leq C(1+\eta)(a^2+b^2). \tag{1.2}$$

An anti-self-dual (ASD) Hodge identity checks the relative coefficients of the two mixings and proves positivity and preservation of the unbroken kernels. The correction relative to the actual leading gauge masses is $O(\eta(a^2+b^2))$, uniformly in b/a. Equation (1.2) concerns the defined pencil; it does not bound an uncomputed remainder in exact string poles.

The interaction result concerns two specified massless bundle scalars. Put $r^2 = 2(a^2+b^2)$ and $e^2 = 2g_{10}^2/(3V_0L^4)$, with g_{10} defined by the ten-dimensional gauge action. Let $\mathcal{M}_{\mathrm{YM}}$ be their on-shell four-point tree amplitude on the actual internal ASD background, including the complete internal Yang–Mills vector and scalar spectrum. For $0 < b \leq \kappa_0 a$, with fixed $\kappa_0 < 1$, we prove

$$\mathcal{M}_{\mathrm{YM}} = e^2 \left(\frac{t-u}{m_0^2 - s} + \frac{t-s}{m_0^2 - u} \right) + O\left(e^2 \left[r^2 \frac{E^2}{m_0^2 + E^2} + (EL)^2 \right] \right), \qquad m_0^2 = \frac{8b^2}{3V_0L^2}. \tag{1.3}$$

Here s, t, u are Mandelstam invariants and $E^2 = \max(|s|, |t|, |u|)$. The domain lies below the two heavier Higgs masses, at a fixed relative distance from the light poles; its precise conditions appear in Theorem 6.2. The error is absolute, with constants independent of b/a. For a sufficiently separated hierarchy the domain contains energies below and above the light mass. It therefore connects the low-energy quotient description with the intermediate description retaining a light U(1) vector multiplet.

Two estimates make this comparison possible. The actual external modes have reference harmonic components accurate to $O(r^2)$ and complementary components of size $O(r)$. The mixed terms vanish in the cubic currents by harmonic orthogonality, giving $O(r^2)$ vertex errors without a bound on derivatives of the balancing gauge. The scalar partners of each vector must also be retained. Their cubic vertices contain their masses, and scalar exchange cancels the zero-momentum contact term exactly. The remaining complementary tower is controlled by a fixed gap and Parseval. These facts yield the complete spectral formula in Proposition 6.1 and the uniform error in (1.3).

The same actual-mode estimates and the Gauss–O'Neill curvature formula give a selected harmonic sectional curvature $K_{g^{\mathrm{KS}}} = 3(1+O(r^2))/(2b^2)$, uniformly in this hierarchy. Solving the mixed Einstein and two-form constraints, and retaining the other neutral fields and their different kinetic scales, identifies the leading physical curvature coefficient as

$$\mathcal{K}_{\mathrm{phys}}^{\mathrm{lead}} = \frac{3g_{10}^2}{4L^2b^2}[1 + O(a^2+b^2)], \qquad m_L^2 \mathcal{K}_{\mathrm{phys}}^{\mathrm{lead}} = 3e^2[1 + O(a^2+b^2)]. \tag{1.4}$$

The identification first extracts the leading tree large-volume coefficient at each regular point; the resulting coefficient obeys the uniform geometric estimate. It does not bound higher physical coefficients in a simultaneous finite-η limit. General metric derivatives can still diverge near a changing stabilizer, as an explicit horizontal projector in appendix B demonstrates.

The first additional heterotic tree coefficient requires the full action through bare order α'^2 after canonical normalization. We give its order count in section 8. The same harmonic energy bound controls quadratic potential mixing with K3 metric deformations and gives a uniform linear normal lift. We also calculate the universal graviton contribution, which is bounded away from forward scattering and has a massless pole absent from the leading Yang–Mills amplitude. These calculations do not determine the full corrected scalar vertices and their combined metric, dilaton, two-form and gauge exchanges. In particular, (1.3) is not a finite-string-scale error estimate for the full heterotic amplitude.

Heterotic bundle stability, charged fields and D-term Higgsing have an established effective-theory description [2]. The SO(32) embedding, spectrum and abelian mass criteria used here follow the K3 analysis of [3]. Bundle and matter metrics were computed in [4], while the general geometry–bundle–two-form fibration and its large-volume hierarchy were developed in [5]. We retain the compensator and curvature-squared terms needed to identify the leading coefficient; we do not import a specialized duality completion as the metric of the present compactification. The relation between scalar curvature and covariant amplitudes is also standard [6]. Kaluza–Klein amplitude cancellations derived from Hodge decomposition and eigenfunction completeness are studied in [7]. Here the required estimates concern covariant modes on a nontrivial ASD bundle whose Higgs scales separate.

Universal heterotic representatives and higher-order moduli metrics are developed in [8–10]; the recursive second-order action is also discussed in [11]. Their Calabi–Yau threefold results concern four-dimensional compactifications; we use the action and covariant-variation conventions, not their moduli-metric theorems as results for six-dimensional K3 vacua.

The quantitative input is the energy-relative spectral and quotient-metric comparison in [1]. We use the local harmonic quadraticity argument of [12] without extending that paper's global compactness theorem to this family. The new comparison concerns the actual scalar representatives and the complete Yang–Mills amplitude uniformly through a separating Higgs hierarchy. The relative vector correction uses the same connection's gauge energy and a harmonic transgression identity; the fixed-ray coefficient and its Pauli selection rule are calculated in appendix A. Classical quotient curvature and local Ricci-flat estimates supply geometric identifications, while the uniform vertex and spectral-sum estimates determine the accuracy in this degenerating family.

Section 2 constructs the family and fixes its local coordinates. Sections 3 and 4 derive the vector statements and the retained field content. Sections 5 and 6 prove the actual-mode estimates and the amplitude comparison. Section 7 identifies the leading ambient scalar interaction, and section 8 treats the first additional tree order. Throughout, the string coupling is taken in the tree regime independently of the geometric parameters.

2 The compactification and its enhancement locus

2.1 Stable factors and the orthogonal embedding

Fix a smooth projective K3 surface S with $\mathrm{Pic}(S) = \mathbb{Z}H$, where H is ample. Let g_0 be its Ricci-flat Kähler metric in that class, oriented so that primitive $(1,1)$ forms are anti-self-dual. Norms and harmonic representatives without an explicit scale use g_0 and unnormalized trace inner products. Write $V_0 = \mathrm{Vol}(S, g_0)$.

Choose locally free μ_H-stable bundles A, B with

$$\begin{aligned} \operatorname{rk} A = \operatorname{rk} B = 3, \qquad & \det A = \det B = \mathcal{O}_S, \\ c_2(A) = 7, \quad c_2(B) = 10, \qquad & A \not\simeq A^*, \quad B \not\simeq B^*. \end{aligned} \tag{2.1}$$

Here existence includes the locally free, slope-stable and non-self-dual conditions. We give the argument because these conditions determine both the analytic application and the unbroken gauge algebra.

For $k = 7, 10$ the Mukai vector is $v_k = (3, 0, 3-k)$, with $v_7^2 = 24$ and $v_{10}^2 = 42$. Both vectors are primitive. An equal reduced-Hilbert-polynomial proper subsheaf would have first Chern class zero and Euler characteristic divided by rank equal to $-1/3$ or $-4/3$. Neither is possible for rank one or two. Thus H is general for these vectors. Yoshioka's nonemptiness and irreducibility theorem, together with the slope-stability and local-freeness criteria in [13, Theorem 0.1, Proposition 0.5 and section 3.3], gives nonempty slope-stable and locally free opens. In that notation the integer l equals three and $v_k^2 \geq 2l^2$. The exceptions to the locally free criterion do not occur: the vectors have rank three, whereas the spherical exception of the form $(\operatorname{rk} v_0)v_0 - \omega$ would have square rank. Irreducibility makes the two opens intersect. Since $\mathrm{Pic}^0(S) = 0$, their determinant is trivial.

Dualization acts holomorphically as an involution on this smooth open. At a fixed point F, simplicity makes an isomorphism $F \to F^*$ symmetric or skew-symmetric after scalar normalization. Odd rank excludes the latter. The invariant tangent space is $H^1(S, \mathfrak{so}(F)) \simeq H^1(S, F)$: use $\Lambda^2 F^* \simeq F$ and the symmetric form. Stability and Serre duality give $H^0(F) = H^2(F) = 0$, and Riemann–Roch gives

$$h^1(F) = k - 6, \qquad h^1(\operatorname{End} F) = 6k - 16. \tag{2.2}$$

The fixed locus of a holomorphic involution is smooth with its invariant tangent space. One can see this directly by replacing local coordinates z centered at a fixed point by $\frac{1}{2}(z + D\iota_0\, z\circ\iota)$; their derivative is the identity and the action becomes linear. The fixed locus therefore cannot fill the stable locally free open. This proves that the choices in (2.1) exist. Their distinct second Chern classes also exclude isomorphisms between A or A^* and B or B^*.

At enhancement set $E_0 = A \otimes \mathbb{C}^2$ and define the real gauge bundle

$$\mathcal{V} = (E_0 \oplus B)_{\mathbb{R}} \oplus \mathbb{R}^{14}, \qquad \mathcal{V} \otimes \mathbb{C} = E_0 \oplus E_0^* \oplus B \oplus B^* \oplus \mathcal{O}^{14}. \tag{2.3}$$

It has real rank 32, $w_2(\mathcal{V}) = 0$, and $p_1(\mathcal{V}) = -2c_2(E_0 \oplus B) = -48 = p_1(TS)$. The weighted instanton number is

$$2c_2(A) + c_2(B) = 2 \cdot 7 + 10 = 24. \tag{2.4}$$

There is consequently no five-brane source in the integrated Bianchi identity [3]. The Donaldson–Uhlenbeck–Yau correspondence [14, 15] supplies determinant-flat HE connections, which are ASD on S. They give the leading background with eight supercharges, or six-dimensional $\mathcal{N} = (1, 0)$ supersymmetry. Equation (2.4) is an integrated condition; it does not set the gauge and tangent curvature densities equal pointwise. Their difference sources the first correction in section 4.

The parallel orthogonal adjoint sections have compact algebra

$$\mathfrak{h}_0 = \mathfrak{so}(14) \oplus \mathfrak{u}(2) \oplus \mathfrak{u}(1), \qquad \dim_{\mathbb{R}} \mathfrak{h}_0 = 96. \tag{2.5}$$

Indeed the endomorphism blocks give $\mathrm{End}(\mathbb{C}^2)$ and $\mathbb{C}$ by simplicity. The additional exterior-square blocks have no parallel sections. For example,

$$\Lambda^2(A \otimes \mathbb{C}^2) = (\Lambda^2 A \otimes \mathrm{Sym}^2 \mathbb{C}^2) \oplus (\mathrm{Sym}^2 A \otimes \Lambda^2 \mathbb{C}^2),$$

and $H^0(A \otimes A) = \mathrm{Hom}(A^*, A) = 0$ by stability and non-self-duality. The other cross blocks vanish for the same reason. There are no maps to the trivial fourteen-dimensional factor. This also explains why replacing A by a pseudoreal rank-two factor would give a different centralizer. All central factors here have $c_1 = 0$; the abelian flux mass matrix of [3, section 3.2] vanishes. Protection of the nonconstant-profile vector calculation requires the separate argument in section 4.

2.2 Harmonic coordinates and actual bundles

General K3 formality results [16] identify quadratic deformation germs. The quantitative metric application below needs the stronger coordinate specification in the original harmonic chart.

The harmonic space $H^1(S, \mathrm{End}\, A)$ has complex dimension 26 and a quaternionic structure. Choose thirteen orthonormal symplectic pairs, normalized using the parallel holomorphic two-form. Then

$$H^1(S, \mathrm{End}\, E_0) = H^1(S, \mathrm{End}\, A) \otimes \mathrm{End}(\mathbb{C}^2)$$

has coordinates $(X_j, Y_j)_{j=1}^{13}$ with the flat real metric

$$\|\dot{x}\|^2 = \mathrm{Re} \sum_{j=1}^{13} \mathrm{tr}(\dot{X}_j^* \dot{X}_j + \dot{Y}_j^* \dot{Y}_j). \tag{2.6}$$

The multiplicity group acts by simultaneous conjugation; its center acts trivially. The complex and real moment equations are

$$\sum_j [X_j, Y_j] = 0, \qquad \sum_j \big([X_j, X_j^*] + [Y_j, Y_j^*]\big) = 0. \tag{2.7}$$

Only their zero level is used. Let $\mathcal{C}$ be this hyperkähler quotient by the effective group PU(2), with metric $g_{\mathcal{C}}$ on its smooth-simple part.

We record why these are exact local deformation coordinates, rather than merely the quadratic part of an unspecified obstruction. Put $d = \bar{\partial}_{E_0}$ and denote harmonic projection and Green operators by H_q, G_q. The convergent harmonic recursion of [1, Lemma 2.1] is

$$\mathfrak{a}(x) = x - d^* G_2(\mathfrak{a}(x) \wedge \mathfrak{a}(x)), \qquad \mathfrak{k}(x) = H_2(\mathfrak{a}(x) \wedge \mathfrak{a}(x)). \tag{2.8}$$

On a Ricci-flat K3 its obstruction is exactly

$$\mathfrak{k}(x) = H_2(x \wedge x). \tag{2.9}$$

For completeness, a harmonic $\operatorname{End} E_0$-valued $(0,2)$-form is $\xi\bar{\sigma}$ with ξ parallel. This follows from the formula for $d^*(\xi\bar{\sigma})$ and the Bochner identity on the zero-mean-curvature endomorphism bundle. If $d^*u = 0$ and $\gamma = \phi\bar{\sigma}$, the coefficients of $u \wedge d^*\gamma$ and $d^*\gamma \wedge u$ are, up to common nonzero scalars, $-\sum_j u_{\bar{j}} \nabla_j \phi$ and $\sum_j (\nabla_j \phi) u_{\bar{j}}$. Pair either with a parallel ξ and integrate by parts, keeping the matrix order. The resulting terms contain $\sum_j \nabla_j u_{\bar{j}} = -d^*u$ and vanish. Consequently both products have zero harmonic projection. Writing $\mathfrak{a} = x + \beta$ with β coexact and d^*-closed eliminates all terms containing β in the obstruction and proves (2.9). This is precisely the local argument in [12, Lemma 3.1]; it uses none of that paper's later global rank or no-bubbling hypotheses.

Serre duality identifies (2.9) with the first equation in (2.7); a common nonzero symplectic factor and the rank-dual identification do not change its zero set. Fixing the determinant removes no harmonic directions because $H^1(S, \mathcal{O}) = 0$. The analytic quotient construction [17] and HE reconstruction [1, Theorem 1.1] therefore identify the balanced quotient with the actual nearby HE moduli. This identification Ψ is holomorphic in these same coordinates: the holomorphic classifying map and the reconstructed HE map assign the same bundle class to each harmonic deformation. No further nonlinear reparametrization is made.

The quantitative input, specialized to this fixed background, is

$$(1 - Cr^2) g_{\mathcal{C}} \;\le\; \Psi^* g^{\mathrm{KS}} \;\le\; (1 + Cr^2) g_{\mathcal{C}}, \qquad r = \|x\|, \tag{2.10}$$

on genuine smooth stratum tangents. Here g^{KS} is the real harmonic Kodaira–Spencer metric in the unnormalized Dolbeault norm. The same constant works when positive orbit scales have arbitrary ratios. It may depend on the fixed metric, bundles, ranks, harmonic dimensions and elliptic gaps. This does not assert uniformly sized smooth coordinate charts through every lower stratum. On the smooth-simple locus g^{KS} is hyperkähler [18].

Let σ_j be the Pauli matrices with $\operatorname{tr}(\sigma_i \sigma_j) = 2\delta_{ij}$ and $[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k$. Our family is

$$X_1 = a\sigma_3, \qquad X_2 = b\sigma_1, \qquad X_j = 0 \ (j \ge 3), \qquad Y_j = 0, \qquad a, b \in \mathbb{R}. \tag{2.11}$$

It satisfies both moment equations since its nonzero matrices are Hermitian. If $ab \neq 0$, a line invariant under σ_3 is an eigenline, and σ_1 exchanges the two eigenlines. The representation is

therefore simple. Reconstruction gives an actual stable rank-six bundle $E_{a,b}$ with $c_2 = 14$ and trivial determinant. The local simple moduli space has complex dimension $104 - 2 \cdot 3 = 98$.

On a nonzero axis the representation is the sum of two distinct characters, corresponding to two nearby stable rank-three factors. Upper semicontinuity preserves the other Hom vanishings after shrinking the neighborhood. The unbroken algebra is thus $\mathfrak{so}(14) \oplus \mathfrak{u}(1)^2$ generically, $\mathfrak{so}(14) \oplus \mathfrak{u}(1)^3$ on an axis, and (2.5) at the origin. Their dimensions are 93, 94 and 96, respectively.

3 Light vectors and the retained spectrum

3.1 The physical vector operator

On real skew-adjoint generators set $B_F(U, V) = -\frac{1}{2} \operatorname{Tr}_{32}(UV)$. Define g_{10} by

$$S_{\mathrm{YM}} = -\frac{1}{4g_{10}^2} \int B_F(F_{MN}, F^{MN})\, d\operatorname{Vol}_{10}\,. \tag{3.1}$$

The constant dilaton is absorbed in this coupling. Initially use the product spacetime metric $g_{1,5} + L^2 g_0$; a constant change to six-dimensional Einstein frame must rescale masses and scalar kinetic tensors consistently. On a realified unitary block B_F restricts to $-\operatorname{Re}\operatorname{Tr}$. A multiplicity generator at E_0 consequently has reference norm $3V_0 \operatorname{tr}_2(\xi^*\xi)$.

For a Lorentz-transverse external fluctuation $\delta\mathcal{A}_\mu = v_\mu \psi(y)$, background gauge separates the internal Goldstone variation. The external kinetic and mass forms are $g_{10}^{-2}\|\psi\|^2_{L^2(L^2 g_0)}$ and $g_{10}^{-2}\|d_{\mathcal{A}}\psi\|^2_{L^2(L^2 g_0)}$. The coupling cancels in their generalized eigenvalue. On endomorphism zero-forms the HE Kähler identity gives

$$d^*_{\mathcal{A}} d_{\mathcal{A}} = 2\Delta_{\bar\partial, \operatorname{End} E}. \tag{3.2}$$

The induced endomorphism mean curvature is zero, so the ∂ and $\bar\partial$ Laplacians agree in this degree. This is a zero-form vector-profile identity, not an identity for the full one-form Yang–Mills Hessian; compare [4, appendix A]. The zero-form and one-form norms scale as L^4 and L^2, respectively, so physical squared masses acquire the factor L^{-2}.

For $\xi = \sum z_j \sigma_j$ the finite Dolbeault energy and mass are

$$q_x(\xi) = \sum_j \|[\xi, X_j]\|^2 + \|[\xi, Y_j]\|^2 = 8\big(a^2|z_1|^2 + (a^2+b^2)|z_2|^2 + b^2|z_3|^2\big), \tag{3.3}$$

$$M_0(\xi) = 3V_0 \operatorname{tr}(\xi^*\xi) = 6V_0 \sum_j |z_j|^2. \tag{3.4}$$

Thus its ordered eigenvalues for $0 < b \le a$ are $\frac{4}{3V_0}(b^2, a^2, a^2+b^2)$.

Proposition 3.1. *For the actual bundles of (2.11), sufficiently near the origin, there are exactly three real vector species whose positive masses tend to zero away from the unbroken kernel. They satisfy (1.1), with a constant independent of b/a. The corresponding two-sided inequality retains exact zeros on the axes and at the origin. All other nonzero vector-profile eigenvalues have a fixed positive lower bound in reference units.*

Proof. Apply [1, Theorem 1.1(ii)] at the locally free polystable bundle $A \otimes \mathbb{C}^2$, at constant polarization and fixed determinant. Its trace-free multiplicity algebra has dimension three. The theorem bounds every small Dolbeault eigenvalue between $(1 - C\|x\|^2)\nu_j(x)$ and $(1 + C\|x\|^2)\nu_j(x)$, including zeros, and gives a positive complementary gap. Here $\|x\|^2 = 2(a^2+b^2)$ and (3.3)–(3.4) give ν_j. Equation (3.2) and the internal scale give (1.1). The real adjoint sector is preserved by this Laplacian. Each complex eigenspace is the complexification of its Hermitian, or equivalently skew-Hermitian, real form. Its complex multiplicity is therefore the number of real generator profiles, not twice that number. The remaining orthogonal-adjoint blocks have the kernels already identified in (2.5); their positive spectrum stays separated from zero by elliptic continuity on the fixed compact background. □

The essential input is the relative form of the imported estimate. In its proof the complementary Schur term is bounded by the finite energy of the same vector. A uniform additive error of order $(a^2 + b^2)^2$ would not establish the result when $b \ll a^2$. There is no additional gauge coupling multiplying the masses in (1.1) after the convention (3.1) is fixed.

3.2 Massless fields and Higgs multiplets

The vector gap does not make the remaining matter heavy. The full orthogonal adjoint decomposes into endomorphism, exterior-square, mixed tensor and fourteen-dimensional vector sectors. For a stable determinant-trivial rank-r bundle F with $c_2(F) = k$ and the Hom vanishings established above, Riemann–Roch and Serre duality give

$$\begin{aligned} h^1(F) &= k - 2r, & \dim_{\mathbb{H}} \mathcal{M}_F &= rk - r^2 + 1, \\ h^1(F \otimes G) &= r_G k_F + r_F k_G - 2r_F r_G, & h^1(\Lambda^2 F) &= (r-2)k - r(r-1). \end{aligned} \tag{3.5}$$

These follow by integrating $\mathrm{ch}(\cdot)\,\mathrm{td}(S)$; the exterior-square formula uses $c_2(\Lambda^2 F) = (r-2)c_2(F)$ when $c_1(F) = 0$. The neutral moduli formula follows from $h^1(\mathrm{End}\, F) = 2rk - 2r^2 + 2$. In the spectrum convention of [3], complex conjugate sectors combine into full hypermultiplets. At a generic point the dimension-weighted counts are listed in table 1.

At enhancement the E endomorphism sector consists of thirteen adjoint-U(2) hypermultiplets and contributes 52. On a nonzero axis the two distinct rank-three factors contribute $2 \cdot 13$ neutral hypermultiplets and 24 off-diagonal charged ones, giving 50. The other dimension-weighted totals are unchanged. Thus

$$(n_H, n_V) = \begin{cases} (340, 96), & a = b = 0, \\ (338, 94), & ab = 0, \quad a^2 + b^2 > 0, \\ (337, 93), & ab \neq 0. \end{cases} \tag{3.6}$$

There is one tensor multiplet, containing the six-dimensional dilaton, and the gravity multiplet. Every row obeys $n_H - n_V + 29 n_T = 273$. This is a gravitational anomaly count; it does not replace the gauge-anomaly and embedding analysis of [3]. As a separate index check, the real gauge ASD complex has $h^1 - h^0 - h^2 = 4 \cdot 30 \cdot 24 - 4 \cdot 496 = 896$ on K3. Our generic

Table 1. Generic massless hypermultiplet contributions. Charges refer to the two central U(1) factors. The fourteen-dimensional sectors contain hypermultiplets in the vector representation of SO(14), not additional gauge vectors.

Sector	Dimension-weighted hypermultiplet count
E moduli, neutral	49
B moduli, neutral	22
$\Lambda^2 E$, charge $(2,0)$	26
$\Lambda^2 B$, charge $(0,2)$	4
$E \otimes B$, charge $(1,1)$	66
$E \otimes B^*$, charge $(1,-1)$	66
E sector in **14**, charge $(1,0)$	28
B sector in **14**, charge $(0,1)$	56
K3 geometry and two-form fields	20
Total	337

gauge hypermultiplets contribute $4(337-20)$, and $h^0 = 93$ gives $4(337-20) - 4 \cdot 93 = 896$; the corresponding enhanced and axis counts agree as well. Here the expression subtracts all four parallel contributions, equivalently using $h^2 = 3h^0$ in the ASD index.

Three hypermultiplets and three vectors combine into three massive $\mathcal{N} = (1,0)$ vector multiplets generically. Each massive vector has five physical polarizations and three physical real scalar partners; the fourth scalar of its hypermultiplet is a Goldstone mode. On an axis only two such multiplets are massive.

3.3 The common low-energy space

Choose background gauge, diffeomorphism gauge and two-form gauge at the reference background. The leading internal operators are the adjoint zero-form Laplacian for vectors, the positive ASD deformation square for connection scalars, the physical Lichnerowicz operator for metric deformations, ordinary Hodge Laplacians for forms, and the corresponding Dirac complexes for fermions. The ASD square is $d_{\mathcal{A}} d_{\mathcal{A}}^* + 2 d_{\mathcal{A}}^{+*} d_{\mathcal{A}}^+$ on one-forms. The parallel-spinor factorization gives nonnegativity of the physical Ricci-flat metric operator [19]; this excludes the unphysical Euclidean conformal mode. Compactness gives finite kernels and positive complementary gaps. Supersymmetry identifies the physical superpartners.

Take a cutoff below the minimum of these reference gaps and retain every reference low mode, including the gauge/Goldstone complex. Smooth elliptic dependence gives common finite low spectral clusters and a complementary gap $\gamma > 0$ for all sufficiently small (a,b). One must not first divide out a different number of scalar gauge directions at each stratum and then assert continuity on a fixed space. The theory with the Higgs multiplets retained has the common leading window $E_{\text{probe}} \ll \sqrt{\gamma}/L$. Eliminating them instead requires $E_{\text{probe}} \ll b/L$

on a generic regular patch. Higher-derivative string terms are used perturbatively through leading resolvents, not as a positive second-order operator on the entire tower.

4 The first coupled vector correction

4.1 Action convention and the corrected background

For this calculation use unit-scale internal coordinates and remove the common gravitational prefactor. Set $\eta = \alpha'/L^2$ and use positive forms $B_F = -\frac{1}{2}\,\mathrm{Tr}_{32}$ and $B_R = -\frac{1}{2}\,\mathrm{Tr}_T$ on the compact gauge and internal Euclidean rotation algebras. The Lorentzian extension of the tangent trace is understood in the ten-dimensional action. In mostly-plus signature our convention is

$$\begin{aligned} S = \int e^{-2\Phi}\sqrt{-G}\Big[R + 4|d\Phi|^2 - \frac{1}{12}H_{MNP}H^{MNP} \\ - \frac{\eta}{4}\big(B_F(F_{MN}, F^{MN}) - B_R(R_{MN}, R^{MN})\big)\Big]. \end{aligned} \tag{4.1}$$

Use the field choice in which no additional independent purely bosonic operator at order α'^2 is required beyond the recursively defined torsion and composite-connection terms [20, 21]. This statement does not discard their order-α'^2 contributions. The tangent connection is the composite Hull connection $\Theta_H = \Theta_{\mathrm{LC}} + \frac{1}{2}H$, where $H^a{}_b = H^a{}_{bM}dx^M$ in an orthonormal frame, in the convention of [10, section 2.1]. It reduces to Levi–Civita at leading order. Fix Chern–Simons signs by

$$dH = \frac{\eta}{2}\big(B_R(R\wedge R) - B_F(F\wedge F)\big), \tag{4.2}$$

$$\delta H = d\mathcal{B} - \eta B_F(\delta\mathcal{A}\wedge F) + \eta B_R(\delta\Theta\wedge R). \tag{4.3}$$

$\mathcal{B}$ denotes the gauge-covariantly shifted two-form variation. These choices fix relative normalizations rather than combining formulas with incompatible trace or dilaton conventions.

For $t = (a, b)$ let F_t be the full leading ASD curvature. Through first order the smooth background is

$$G_S = (1 + \eta f_t)g_0, \qquad \Phi = \Phi_0 + \tfrac{1}{2}\eta f_t, \qquad H = -\eta *_0 df_t, \tag{4.4}$$

where f_t has zero mean and solves

$$\nabla^m\nabla_m f_t = \frac{1}{4}\big(B_R(R_{mn}, R^{mn}) - B_F(F_{t,mn}, F_t^{mn})\big). \tag{4.5}$$

Here ∇^2 has the divergence sign; the nonnegative scalar Laplacian is $\Delta_0 = -\nabla^2$. The source has mean zero by (2.4). Scalar elliptic inversion gives a unique solution, with uniform spatial bounds on a sufficiently small family of reconstructed connections. These connections have uniformly bounded spatial coefficients in fixed gauge even though the small spectral ratios degenerate. Parameter derivative bounds used later are only on fixed regular patches. For sufficiently small η, $1 + \eta f_t > 0$.

The normalization can be checked directly. In four dimensions the linear conformal Ricci tensor is $-\nabla_m\nabla_n f - \frac{1}{2}g_{mn}\nabla^2 f$; the dilaton Hessian adds $\nabla_m\nabla_n f$. For an ASD curvature, $B_F(F_{mp}, F_n{}^p) = \frac{1}{4}g_{mn}B_F(F_{pq}, F^{pq})$. The Einstein equation from (4.1) gives (4.5). Also $B_F(F\wedge F) = -\frac{1}{2}B_F(F_{mn}, F^{mn})dV$ and $d(-*df) = -\nabla^2 f\, dV$, so the Bianchi identity gives the same equation. The dilaton equation gives the same trace condition, and the two-form equation holds because the leading weighted dual flux is exact. The gauge equation $D_{\mathcal{A}}(e^{-2\Phi}*_4 F) - e^{-2\Phi}F\wedge *_4 H = 0$ also cancels through this order, using $*_4 F = -F$ and $*_4 H = \eta df$. The corresponding conformal/dilaton ansatz is the usual first-order supersymmetric K3 construction [22]; ASD persists under conformal rescaling. It is a solution through the computed order, not an existence assertion for an exact finite-η background.

4.2 The two heavy spin-one sectors

Let P_t be the whole 96-dimensional low real spectral projection of $d^*_{\mathcal{A}_t}d_{\mathcal{A}_t}$ on the full orthogonal adjoint. Put $A_t = P_t d^*_{\mathcal{A}_t}d_{\mathcal{A}_t}P_t$. This space includes all persistent massless vectors. Define

$$T_t p = B_F(p, F_t) \in \Omega^2(S), \qquad (J_t p)_m = B_F(F_{t,mn}, D^n p), \qquad I_t v = \iota_v F_t. \tag{4.6}$$

The Yang–Mills equation and invariance of the trace give $J_t = d^*T_t = I_t^* d_{\mathcal{A}_t}$, and hence $d^*J_t = 0$. In components the latter follows by differentiating $B_F(F_{mn}, D^n p)$; the curvature commutator contracts to zero.

Set $s = \sqrt{\eta}$ and canonically normalize the gauge fluctuation by $p = s\,\delta\mathcal{A}_\mu$. Let v be the metric vector in the coframe $dy^m + v^m_\mu dx^\mu$, and let β be the shifted two-form vector. After the diffeomorphism and form constraints their profiles are co-closed real one-forms. At leading order both have ordinary L^2 kinetic norm and squared-mass operator $\Delta_1 = d^*d$ on co-closed one-forms. The external field-strength coefficient is $-1/4$ for each. For the two-form, the three placements of the internal index in $H_{\mu\nu m}$ turn its contribution to $-H^2/12$ into $-H_{\mu\nu m}H^{\mu\nu m}/4$, giving the external vector kinetic term. The three placements of the external index in $H_{\mu mn}$ instead give its mass term; the factor $1/2$ in the internal two-form norm yields the usual Proca normalization. For the metric the kinetic coefficient follows from the quadratic Kaluza–Klein curvature formula; on a Ricci-flat base its Killing-operator form reduces, on co-closed forms, to $\|dv\|^2$. K3 has no harmonic one-forms, and a Killing field on a compact Ricci-flat K3 would be parallel. Thus both sectors have a fixed positive gap.

With $I_t v = \iota_v F_t$ and the plus coframe just specified, the horizontal mixed gauge curvature is $-d_{\mathcal{A}}\delta\mathcal{A}_\mu - I_t v_\mu$. The terms determining the first low-vector coefficient are therefore

$$Q = \|d_{\mathcal{A}}p + sI_t v\|^2 + \|dv\|^2 + \|d\beta - sT_t p\|^2, \qquad K = \|p\|^2 + \|v\|^2 + \|\beta\|^2 \tag{4.7}$$

at leading heavy order. Reversing the coframe convention reverses the metric mixing sign and leaves its Schur term unchanged. Exact one-form gauge directions are not additional transverse vector species. Harmonic two-form components of $T_t p$, in contrast, must remain in the direct norm in (4.7).

The conformal background contributes gauge kinetic weight $(1+\eta f_t)$: its four-dimensional volume factor and dilaton factor leave one power of $1+\eta f_t$. The extra inverse internal metric in the gauge mass term cancels that power to first order. Thus the direct gauge correction to the pencil is $T_t^* T_t - \lambda f_t$.

Eliminating both complete leading heavy towers gives

$$E_t(\lambda) = P_t \left[T_t^* T_t - \lambda f_t - 2 J_t^* (\Delta_1 - \lambda)^{-1} J_t \right] P_t. \tag{4.8}$$

Each factor of two in this expression comes from one of the two distinct, equally normalized heavy sectors. No finite KK cutoff is taken. Leading elliptic estimates give $\|p\|_{H^k} \le C_k \|p\|$ for $p \in \operatorname{im} P_t$, uniformly in t. Together with the one-form gap and the uniform Poisson bound, this proves, on a fixed interval below half that gap,

$$\|E_t(\lambda)\| + \|\partial_\lambda E_t(\lambda)\| \le C. \tag{4.9}$$

There is no inverse small gauge eigenvalue in this bound.

For clarity, the order count also covers the terms not visible in (4.7). Metric–two-form heavy mixing from the background H and first-order changes of their operators are of order η. Between two gauge–heavy mixings of order $\sqrt{\eta}$ they first affect the low pencil at order η^2. The $\eta \|I_t v\|^2$ term has the same property. A gauge-vector induced torsion variation is of order $\sqrt{\eta}$ in canonical variables, so its quadratic contribution through $\eta R(\Theta)^2$ starts at order η^2. Quadratic gauge Chern–Simons terms have at least two external legs and cannot contract with the purely internal background to supply an earlier diagonal term. Lorentz Chern–Simons corrections to a gauge–heavy mixing start at order $\eta^{3/2}$. Finally the off-diagonal coupling to the gapped gauge complement starts at order η, and its Schur contribution at order η^2. The absence of a further independent bosonic α'^2 operator in the chosen scheme matters here: canonical gauge normalization removes one power of η from a quadratic gauge term. We extract a perturbative coefficient; we do not treat the raw curvature-squared action as a positive bounded perturbation of an infinite-dimensional second-order form.

4.3 Hodge cancellation, positivity and the error bound

Let H_2 denote harmonic projection on real two-forms. For every adjoint zero-form p, ASD gives $*T_t p = -T_t p$. Decompose $T_t p = h + e + c$ into harmonic, exact and coexact parts. Hodge star preserves h and exchanges e, c isometrically, so $\|e\| = \|c\|$. Moreover $e = d\Delta_1^{-1} d^* T_t p$. Hence

$$\|T_t p\|^2 - 2\langle J_t p, \Delta_1^{-1} J_t p \rangle = \|\mathsf{H}_2 T_t p\|^2. \tag{4.10}$$

This elementary identity is a useful check on the two heavy normalizations. Polarization and the resolvent identity rewrite (4.8) as

$$\begin{aligned} E_t(\lambda) &= P_t T_t^* \mathsf{H}_2 T_t P_t - \lambda Z_t(\lambda), \\ Z_t(\lambda) &= P_t \left[f_t + 2 J_t^* \Delta_1^{-1} (\Delta_1 - \lambda)^{-1} J_t \right] P_t. \end{aligned} \tag{4.11}$$

Define the first-order effective model by

$$F_t^{[1]}(\lambda,\eta) = A_t - \lambda I + \eta E_t(\lambda), \qquad m^{2,[1]} = L^{-2}\lambda. \tag{4.12}$$

For small η, its zero-energy mass form is nonnegative, and its kinetic form $I + \eta Z_t(\lambda)$ is positive on the fixed low interval. Also $\partial_\lambda F_t^{[1]} = -I + O(\eta)$ is negative definite there. These assertions concern a finite 96-dimensional pencil with an exactly specified coefficient.

Every parallel generator of the unbroken algebra has $d_{\mathcal{A}_t} p = 0$ and $T_t p = 0$ pointwise. In the central unitary directions the latter follows from determinant-flat curvature on each stable factor; it also holds in the trivial SO(14) block. On an axis the separate rank-three determinants remain trivial because $\mathrm{Pic}^0(S) = 0$. Consequently

$$\ker F_t^{[1]}(0,\eta) = \ker A_t. \tag{4.13}$$

There are exactly 93, 94 or 96 zero roots on the corresponding strata. Positivity precludes an extra kernel outside $\ker A_t$.

4.4 A relative estimate uniform in the Higgs hierarchy

The absolute estimate (4.9) can be sharpened by using the fixed reducible connection $\mathcal{A}_0$. Put $D_0 = d_{\mathcal{A}_0}$, let Π_0 project onto its full 96-dimensional parallel algebra, and set $Q_0 = I - \Pi_0$. The reconstruction in [1, Lemmas 2.1–2.2 and section 3.2] gives a fixed-unitary representative differing from $\mathcal{A}_0$ by $O(r)$ in every fixed spatial regularity norm, where $r^2 = 2(a^2 + b^2)$. Impose the reference Coulomb condition

$$\mathcal{A}_t = \mathcal{A}_0 + \alpha_t, \qquad D_0^* \alpha_t = 0, \qquad \|\alpha_t\|_{C^1} \le Cr. \tag{4.14}$$

This uses only a fixed inverse. Indeed the derivative of $D_0^*(e^u \cdot (\mathcal{A}_0 + \beta) - \mathcal{A}_0)$ in $u \in Q_0 W^{s+1,2}$ is $\pm D_0^* D_0$, with range $Q_0 W^{s-1,2}$, for an integer $s \ge 4$. The implicit-function theorem supplies the small gauge while leaving the stabilizer intact; see also [23, Proposition 3.4]. The following decomposition estimates use this slice; the spectral conclusion is gauge invariant.

Write $D = D_0 + [\alpha_t, \cdot]$ and split a zero-form as $p = k + q$, where $k = \Pi_0 p$. Since $D_0^*[\alpha_t, k] = 0$, the terms $D_0 q$ and $[\alpha_t, k]$ are orthogonal. The fixed Poincaré inequality on Q_0 and absorption of $[\alpha_t, q]$ give

$$\|D_0 q\| + \|[\alpha_t, k]\| + \|q\|_{W^{1,2}} \le C\|Dp\|. \tag{4.15}$$

For example, if $U^2 = \|D_0 q\|^2 + \|[\alpha_t, k]\|^2$ and γ_0 is the first positive reference eigenvalue, then $U \le \|Dp\| + Cr\gamma_0^{-1/2} U$.

On the whole low cluster this improves to

$$\|Q_0 p\| \le Cr\|Dp\|, \qquad p \in \operatorname{im} P_t. \tag{4.16}$$

To prove it first take an eigenvector with eigenvalue λ. The complementary operator $L_Q = Q_0 D^* D Q_0$ has a fixed positive gap, whereas the 96 low eigenvalues are $O(r^2)$ by min–max. The complementary equation is

$$q = -(L_Q - \lambda)^{-1} Q_0 [\alpha_t, \cdot]^* [\alpha_t, k].$$

Equations (4.15) and (4.14) prove (4.16) for that eigenvector. For $p = \sum_{j=1}^{96} c_j e_j$, Cauchy–Schwarz gives $\|Q_0 p\| \le Cr\sqrt{96}(\sum_j |c_j|^2\lambda_j)^{1/2}$. Absorb the fixed dimension into C. Zero eigenvalues are included.

Every reference parallel k has $B_F(k, F_0) = 0$ pointwise. On $A \otimes \mathbb{C}^2$ this follows from $F_0 = F_A \otimes I$ and $\operatorname{tr} F_A = 0$; it also holds on the determinant-flat B block and the trivial block. The curvature identity therefore gives the exact transgression formula

$$\mathsf{H}_2 B_F(p, F_t) = \mathsf{H}_2 B_F(q, F_t) + \mathsf{H}_2 B_F(k, \alpha_t \wedge \alpha_t). \tag{4.17}$$

The omitted linear term is $dB_F(k, \alpha_t)$ and has zero harmonic projection. Trace invariance bounds the second term by $C\|\alpha_t\|_\infty \|[\alpha_t, k]\|$. Equations (4.15)–(4.17) imply

$$0 \le M_t := P_t T_t^* \mathsf{H}_2 T_t P_t \le Cr^2 A_t. \tag{4.18}$$

The vanishing reference harmonic flux is essential: a nonzero $\mathsf{H}_2 B_F(k, F_0)$ would violate this inequality at $r = 0$.

The zero-mean convention in (4.5) also matters. Since $B_F(k, k)$ is constant on K3, $\langle k, f_t k\rangle = 0$. Equations (4.16) and the $O(r^2)$ low spectral upper bound yield $\|P_t f_t P_t\| \le Cr^2$. Moreover $\|J_t p\| \le C\|Dp\|$, so on a fixed interval $|\lambda| \le \Lambda_*$ below the one-form gap,

$$\|Z_t(\lambda)\| \le Cr^2, \qquad \partial_\lambda F_t^{[1]} = -I - \eta P_t f_t P_t - 2\eta P_t J_t^* (\Delta_1 - \lambda)^{-2} J_t P_t = -I + O(\eta r^2). \tag{4.19}$$

Proposition 4.1. *There are $r_0, \eta_0, C > 0$, depending only on the fixed compactification data, such that for $r \le r_0$ and $0 \le \eta \le \eta_0$ the pencil (4.12) has 96 real low roots, counted with multiplicity. If μ_j are the ordered eigenvalues of A_t, then*

$$(1 - C\eta r^2)\mu_j \le \lambda_j^{[1]} \le (1 + C\eta r^2)\mu_j, \qquad 1 \le j \le 96. \tag{4.20}$$

The estimate includes exact zeros and has a constant independent of the ratio b/a. The three positive generic masses consequently satisfy (1.2).

Proof. Equations (4.18) and (4.19) give

$$A_t \le A_t + \eta M_t \le (1 + C\eta r^2)A_t, \qquad (1 - C\eta r^2)I \le I + \eta Z_t(\lambda) \le (1 + C\eta r^2)I.$$

The pencil decreases strictly on the fixed interval and has opposite definite signs at its endpoints after shrinking r_0 and η_0. Each ordered eigenvalue function has one root. The min–max principle bounds the jth root by

$$\frac{\mu_j}{1 + C\eta r^2} \le \lambda_j^{[1]} \le \frac{(1 + C\eta r^2)\mu_j}{1 - C\eta r^2}.$$

This proves (4.20), including zero branches. Combining with Proposition 3.1 gives the stated comparison with the finite masses. No positive eigenvalue is replaced by the smallest one in this argument. □

The formal full expression is $F = A_t - \lambda I + \eta E_t(\lambda) + O_{\text{formal}}(\eta^2)$. Here O_{formal} records uncomputed higher coefficients; it is not a uniform numerical bound on an exact finite-η pole. Solving (4.12) retains its energy dependence and thereby resums some higher powers by convention. Those powers are not additional calculated string coefficients. The relative model estimate permits fixed sufficiently small η as b/a tends to zero. Control of higher physical coefficients is a separate question.

5 Actual scalar modes and their gauge vertices

We now retain the light vector multiplets and calculate their interactions with two massless bundle scalars. All fields in this section belong to the actual ASD connection $\mathcal{A} = \mathcal{A}(a, b)$. Fix $0 < \kappa_0 < 1$ and restrict to

$$0 < b \le \kappa_0 a, \qquad r^2 = 2(a^2 + b^2). \tag{5.1}$$

Constants may depend on κ_0 and the fixed compactification, but not on b/a. Sobolev norms below concern spatial derivatives on the fixed K3; they do not denote derivatives with respect to a or b.

5.1 Harmonic representatives of the external states

In the harmonic coordinates of section 2.2, choose the two Dolbeault-unit vectors

$$\beta_u : \ \delta X_3 = \sigma_1/\sqrt{2}, \qquad \beta_v : \ \delta X_3 = \sigma_2/\sqrt{2}, \tag{5.2}$$

with all other components zero. The affine Hermitian X_3 plane satisfies all three moment equations. Its tangents are perpendicular to the gauge orbit and its quaternionic images at the Pauli point. The corresponding real-unit one-forms at the reference connection are

$$\mathsf{u}_0 = (\beta_u - \beta_u^*)/\sqrt{2}, \qquad \mathsf{v}_0 = (\beta_v - \beta_v^*)/\sqrt{2}. \tag{5.3}$$

Transport their classes by Ψ, take actual ASD harmonic representatives, and normalize the resulting ordered pair by the positive inverse square root of its 2×2 Gram matrix. Denote these real-unit external states by u, v. This fixes their frame as well as their plane. Let Π_1 be reference ASD harmonic projection on real one-forms.

Lemma 5.1. *In the fixed-reference unitary Coulomb presentation, for every fixed spatial Sobolev order s,*

$$\begin{aligned} \|\Pi_1 \mathsf{u} - \mathsf{u}_0\|_{H^s} + \|\Pi_1 \mathsf{v} - \mathsf{v}_0\|_{H^s} &\le C_s r^2, \\ \|(1 - \Pi_1)\mathsf{u}\|_{H^s} + \|(1 - \Pi_1)\mathsf{v}\|_{H^s} &\le C_s r. \end{aligned} \tag{5.4}$$

These estimates are uniform on (5.1).

Proof. The fixed-complement reconstruction of [1, Lemmas 2.1–2.2, Proposition 2.3 and Lemma 3.1] has a holomorphic map $\mathfrak{a}(y)$, its derivative J_y, and a complementary Hermitian correction $\sigma(y)$ satisfying

$$\mathfrak{a}(y) = y + O(r^2), \quad J_y = I + O(r), \quad H_1 J_y = I, \quad \bar\partial_0^* J_y = 0, \quad \sigma(y) = O(r^2). \tag{5.5}$$

Here H_1 is Dolbeault harmonic projection; the estimates hold in each fixed spatial norm. Finite balancing replaces x by $y = e^S x$, where $\|S\| \le Cr^2$, and hence $y - x = O(r^3)$. Conjugation acts only on multiplicity matrices. Thus y is still supported in slots X_1, X_2, whereas $w = e^S \beta_u$ or $e^S \beta_v$ is supported in X_3. In particular, $w = \beta_{u,v} + O(r^2)$ and w is flat-orthogonal to the entire complex orbit $O_y = \operatorname{im} C_y$ of the parallel gauge algebra. The exact complex moment equation makes w an integrable tangent.

Minimizing first over the fixed complementary complex gauge space produces the positive Hermitian form B_y on harmonic vectors, with $\|B_y - I\| \le Cr^2$ [1, Proposition 2.4]. The remaining full gauge minimum is the B_y-distance to O_y. If o is the B_y-projection of w onto O_y, then

$$(1 - Cr^2)|o|^2 \le B_y(o,o) = B_y(w,o) = (B_y - I)(w,o), \qquad |o| \le Cr^2 |w|. \tag{5.6}$$

This estimates the orbit vector, without estimating a Lie algebra element whose action gives that vector. No inverse small orbit eigenvalue enters.

For $w - o$, the complementary minimizer ζ satisfies $\|\zeta\|_{H^{s+1}} \le C_s r |w - o|$ by the fixed-gap normal equation and elliptic regularity. Its harmonic representative in the holomorphic frame is

$$\gamma = J_y(w - o) - D_{\mathfrak{a}} \zeta, \qquad D_{\mathfrak{a}} = \bar\partial_0 + [\mathfrak{a}, \cdot].$$

It is closed by integrability and orthogonal to the full complex gauge image for the HE metric, so it is Dolbeault harmonic for that metric. Since $H_1 J_y = I$ and $H_1 \bar\partial_0 \zeta = 0$,

$$H_1 \gamma = w - o - H_1[\mathfrak{a}, \zeta] = \beta_{u,v} + O(r^2).$$

The complementary component is $O(r)$ in H^s.

Transport by e^σ to the reference unitary bundle changes these estimates only by $O(r^2)$. Differentiating this frame along a parameter, or differentiating S, adds a complex gauge direction, removed by the full harmonic minimum. No bound for either parameter derivative is used. Passing to the real representatives (5.3), harmonic orthogonality gives Gram matrix $I + O(r^2)$. Its inverse square root preserves the estimates. Finally the fixed Coulomb correction is generated by a complementary unitary gauge of size $O(r^2)$: the first connection jet is already co-closed, and the reference complementary gauge Laplacian has a fixed gap. Its conjugation changes these one-forms by $O(r^2)$. □

5.2 The light eigenvector

Put $D = d_{\mathcal{A}}$ and $\mathcal{L} = D^* D$. At the reference point let

$$k_i = I_A \otimes i\sigma_i / \sqrt{6V_0}, \qquad i = 1, 2, 3, \tag{5.7}$$

extended by zero on the other orthogonal-adjoint blocks. These are orthonormal. The other 93 parallel sections form an exact fixed kernel throughout the family with $a, b > 0$.

Lemma 5.2. *The normalized light eigenvector p_L of $\mathcal{L}$ can be chosen with $\|p_L - k_3\|_{H^s} \le C_s r^2$. The two heavier low eigenvectors span a plane within $O(r^2)$ of $\operatorname{span}\{k_1, k_2\}$, uniformly on (5.1).*

Proof. Write $\mathcal{A} = \mathcal{A}_0 + \alpha$ in fixed Coulomb gauge. The quadratic term in its unitary reconstruction is reference-harmonic orthogonal: its $(0,1)$ part is $\mathfrak{a}_2 - \bar{\partial}_0 \sigma_2$, with $\mathfrak{a}_2$ coexact. Finite balancing changes x only by $O(r^3)$. The complementary Coulomb correction has size $O(r^2)$; its linear exact term has zero harmonic projection, and its commutator with the first jet has size $O(r^3)$. Consequently

$$\Pi_1 \alpha = \alpha_1(x) + O(r^3), \qquad (1 - \Pi_1)\alpha = O(r^2). \tag{5.8}$$

These are value estimates in fixed spatial norms.

Let Π_0 project onto the full reference parallel algebra. Coulomb orthogonality gives $Q_0 \mathcal{L} \Pi_0 = Q_0 [\alpha, \cdot]^* [\alpha, \cdot] \Pi_0 = O(r^2)$. The fixed complementary gap, or the eigenvector equation used in (4.16), represents the low cluster as a graph $X : \operatorname{im} \Pi_0 \to \operatorname{im} Q_0$ with $X = O(r^2)$ in each fixed spatial norm. Its canonical isometry is $U = (I + X)(I + X^* X)^{-1/2}$. Parallel commutators preserve reference harmonic one-forms and their orthogonal complement. Equation (5.8) therefore gives

$$U^* \mathcal{L} U = \Lambda_x + O(r^4), \qquad \Lambda_x\big|_{\langle k_1, k_2, k_3 \rangle} = \frac{8}{3V_0} \operatorname{diag}(a^2, a^2 + b^2, b^2). \tag{5.9}$$

Indeed the harmonic first-jet/quadratic-jet cross term vanishes, the remaining diagonal error is $O(r^4)$, and the cross block times X and the complementary graph energy have the same order. The fixed 93-dimensional kernel decouples exactly. In its orthogonal three-dimensional complement the light eigenvalue of Λ_x has separation at least $c_{\kappa_0} a^2$ from the other two. The error in its unit eigenvector is bounded by $C r^4 / a^2 \le C r^2$. The orthogonal plane and spatial estimates follow from the graph construction. □

The additive matrix estimate (5.9) does not prove a relative bound for the light mass when $b \ll a^2$. That bound remains the separate energy-relative conclusion of Proposition 3.1.

5.3 Uniform cubic vertices

Let I_α, $\alpha = 1, 2, 3$, be the parallel complex structures on real one-forms. For an adjoint zero-form p define

$$j_p = \langle [p, \mathsf{u}], \mathsf{v} \rangle, \qquad b_p^\alpha(w, z) = \langle I_\alpha [p, w], z \rangle, \quad w, z \in \{\mathsf{u}, \mathsf{v}\}. \tag{5.10}$$

The inner product uses g_0 and B_F. The second form is symmetric because I_α and $\operatorname{ad} p$ are commuting skew-adjoint operators. These are respectively the vector current and the Hessian of a component of the gauge moment map, with signs fixed in section 6.

Proposition 5.3. *For the three low positive vector profiles,*

$$j_{p_L} = -\frac{2}{\sqrt{6V_0}} + O(r^2), \qquad j_{p_M}, j_{p_H} = O(r^2), \qquad b^{\alpha}_{p_j}(w,z) = O(r^2). \tag{5.11}$$

Every persistent zero mode has both vertices exactly zero. For a complete orthonormal zero-form eigenbasis $\{p_n\}$*,*

$$\sum_n \left[|j_{p_n}|^2 + \sum_{\alpha=1}^{3} \sum_{w,z\in\{\mathsf{u},\mathsf{v}\}} |b^{\alpha}_{p_n}(w,z)|^2 \right] \le C. \tag{5.12}$$

All constants are uniform on (5.1)*.*

Proof. For a parallel k, both $\operatorname{ad} k$ and $I_\alpha \operatorname{ad} k$ preserve reference harmonic one-forms and their orthogonal complement. Thus mixed harmonic/complementary pairings vanish in (5.10). Lemma 5.1 leaves an $O(r^2)$ error: the harmonic parts change by $O(r^2)$ and both complementary parts are $O(r)$. Lemma 5.2 and spatial multiplication estimates allow replacement of each low profile by its parallel approximation, with the same error.

At the reference plane, Pauli commutators give $j_{k_3} = -2/\sqrt{6V_0}$ and $j_{k_1} = j_{k_2} = 0$. The three flat moment maps vanish identically on the Hermitian X_3 plane. Their Hessians on each selected pair therefore vanish for every parallel generator. This proves (5.11). The persistent central and spectator generators commute pointwise with the selected $\operatorname{End} E$ fluctuations, proving the exact zero statements.

Each functional in (5.10) is the L^2 pairing of p with a pointwise bilinear zero-form expression in u,v. Lemma 5.1 and Sobolev multiplication bound its L^2 norm uniformly. Parseval gives (5.12), including all complementary SO(32) blocks. □

6 Scattering across the light-vector scale

Let $\mathcal{M}_{\mathrm{YM}}$ be the tree amplitude for $\phi\chi \to \phi\chi$, where ϕ, χ are the canonical massless scalars with internal states u,v. It is defined by the leading ten-dimensional action (3.1) on $\mathbb{R}^{1,5} \times (S, L^2 g_0)$, expanded about the actual ASD connection, with its entire internal gauge and connection-scalar spectrum retained. We use mostly-plus Lorentzian signature, with diagrams summing to $i\mathcal{M}_{\mathrm{YM}}$, and define $s = -(p_1+p_2)^2$, $t = -(p_1-p_3)^2$, $u = -(p_1-p_4)^2$ for the incoming momenta p_1, p_2 and outgoing momenta p_3, p_4. Thus $s+t+u = 0$, with physical $s > 0$. This amplitude is the leading Yang–Mills contribution to the large-volume tree expansion. Additional heterotic interactions are discussed separately in section 8.

The useful normalization and mass scales are

$$g_* = \frac{g_{10}}{L^2}, \qquad e^2 = \frac{2g_*^2}{3V_0}, \qquad m_0^2 = \frac{8b^2}{3V_0L^2}, \qquad M_0^2 = \frac{8a^2}{3V_0L^2}. \tag{6.1}$$

A real-unit internal scalar enters the connection with coefficient g_{10}/L, and a unit zero-form vector with coefficient g_{10}/L^2. Thus g_* is the common gauge cubic factor; the Pauli current in (5.11) supplies the charge e/g_*.

6.1 Scalar partners and the complete connection spectrum

Let $\{p_n\}$ be an orthonormal real eigenbasis of $\mathcal{L} = D^*D$, with eigenvalues $\mu_n \geq 0$ and physical masses $m_n^2 = \mu_n/L^2$. Choose the moment maps $\mu_\alpha(\mathcal{A})$ with signs such that

$$\operatorname{grad}\langle \mu_\alpha, p\rangle = -I_\alpha D p, \qquad D^2\langle \mu_\alpha, p\rangle(w,z) = b_p^\alpha(w,z). \tag{6.2}$$

The notation $\mu_\alpha(\mathcal{A})$ denotes a moment map, whereas μ_n denotes a zero-form eigenvalue. A parallel self-dual Kähler triple has $|\omega_\alpha|^2 = 2$. The curvature contractions defining the moment maps consequently satisfy $\|F^+\|^2 = \frac{1}{2}\sum_\alpha \|\mu_\alpha\|^2$. Subtracting the fixed ASD topological energy from the internal Yang–Mills energy gives

$$V(\mathcal{A}) = \frac{1}{2g_{10}^2}\sum_{\alpha=1}^{3}\|\mu_\alpha(\mathcal{A})\|^2. \tag{6.3}$$

This fixes the scalar-potential normalization.

For $\mu_n > 0$ set

$$n_{n\alpha} = -I_\alpha D p_n/\sqrt{\mu_n}. \tag{6.4}$$

These are orthonormal transverse scalar modes of mass m_n. Together with the Goldstones $Dp_n/\sqrt{\mu_n}$ and the ASD harmonic one-forms, they give the full real connection-one-form spectrum. To verify the assertion, ASD moment-map equivariance gives $D^*I_\alpha D = 0$. Quaternionic multiplication then gives

$$(I_\alpha D)^*(I_\beta D) = \delta_{\alpha\beta}\mathcal{L}, \qquad D\mu_\beta(n_{n\alpha}) = \delta_{\alpha\beta}\sqrt{\mu_n}\, p_n. \tag{6.5}$$

The elliptic ASD deformation operator $D^* \oplus \sqrt{2}D^+$ has kernel the real ASD harmonic space. Identify self-dual forms with three zero-forms using $\omega_\alpha/\sqrt{2}$. Its adjoint components are, up to fixed signs, D, I_1D, I_2D, I_3D. The Gram identities (6.5) identify its product with its adjoint with four copies of $\mathcal{L}$. Its closed-range decomposition proves completeness of the displayed spectrum. The Hessian of (6.3) on $n_{n\alpha}$ is μ_n/g_{10}^2; the kinetic coefficient is L^2/g_{10}^2. Their ratio is m_n^2, as required by the massive vector multiplet.

6.2 The spectral amplitude and contact cancellation

Write $j_n = j_{p_n}$ and $b_{n\alpha} = b_{p_n}^\alpha$. Persistent zero modes have zero vertices by Proposition 5.3. The following sum therefore runs over $\mu_n > 0$.

Proposition 6.1. *Below the fixed complementary Kaluza–Klein gap and away from tree poles, the complete leading Yang–Mills amplitude is*

$$\begin{aligned}\mathcal{M}_{\mathrm{YM}} = g_*^2\sum_{n>0}\Bigg\{ & j_n^2\left(\frac{t-u}{m_n^2-s}+\frac{t-s}{m_n^2-u}\right) \\ & +\sum_{\alpha=1}^{3}\left[b_{n\alpha}(\mathsf{u},\mathsf{v})^2\left(\frac{s}{m_n^2-s}+\frac{u}{m_n^2-u}\right)+b_{n\alpha}(\mathsf{u},\mathsf{u})b_{n\alpha}(\mathsf{v},\mathsf{v})\frac{t}{m_n^2-t}\right]\Bigg\}.\end{aligned} \tag{6.6}$$

The complementary sums converge absolutely.

Proof. Use the linear slice $D^*h = 0$ at the actual background. It removes Goldstones and the quadratic vector/scalar mixing. The scalar kinetic term is flat in h, so there are no pure-scalar derivative cubic vertices. The vector cubic is $-g_* j_n A_\mu(\phi\partial^\mu\chi - \chi\partial^\mu\phi)$. The external current is conserved on shell. Contracting it with the transverse vector propagator in the s and u channels gives the first line of (6.6).

The moment map is quadratic on the affine space of connections. Its first derivative on u, v vanishes. Differentiating (6.3) three times, using (6.5), and inserting the three canonical scalar factors gives the scalar cubic

$$g_* m_n b_{n\alpha}(w, z), \qquad w, z \in \{\mathsf{u}, \mathsf{v}\}. \tag{6.7}$$

All cubic potential vertices among three ASD harmonic modes vanish. For one pairing $(w, z), (w', z')$ of four external states, the quartic potential contributes

$$-g_*^2 \sum_{n,\alpha} b_{n\alpha}(w, z) b_{n\alpha}(w', z')$$

to the invariant amplitude. This is zero-form Parseval applied to $D^2\mu_\alpha$; its zero-eigenspace component vanishes by neutrality. Exchange of the corresponding normal scalar in channel q^2 contributes the same product times $g_*^2 m_n^2/(m_n^2 - q^2)$. Their sum is

$$g_*^2 \sum_{n,\alpha} b_{n\alpha}(w, z) b_{n\alpha}(w', z') \frac{q^2}{m_n^2 - q^2}. \tag{6.8}$$

The three pairings, with $q^2 = s, t, u$, give the second line of (6.6). The exact subtraction removes its zero-momentum term before any small-mass estimate is taken.

A connected four-scalar tree consists of one quartic vertex or two cubic vertices. The preceding list exhausts these in the Yang–Mills action. A vector seagull cannot give such a tree. Fermionic or ghost internal lines, with no external fermions or ghosts, would form loops. Completeness of (6.4) rules out a missing positive connection-scalar sector. There is no massless gauge or scalar exchange for the selected states.

For complementary modes $\mu_n \geq \gamma > 0$ and $|q^2|L^2 \leq \gamma/2$, their denominators are at least $\mu_n/(2L^2)$. The complete Parseval bounds (5.12) and Cauchy–Schwarz prove absolute convergence, both of the contact/exchange recombination and of the remaining vector sum. No finite truncation is required. □

6.3 A comparison uniform in the Higgs hierarchy

Choose $\kappa_0^2 < \theta_H < 1$ and $0 < \delta < 1$, and put $E^2 = \max(|s|, |t|, |u|)$. The comparison domain is

$$0 < b \leq \kappa_0 a, \qquad E^2 \leq \theta_H M_0^2, \qquad |m_0^2 - q^2| \geq \delta(m_0^2 + E^2) \quad (q^2 = s, t, u). \tag{6.9}$$

It lies below the two heavier Higgs masses and stays a fixed relative distance from the light poles and exceptional kinematics. Define

$$\mathcal{M}_0 = e^2 \left(\frac{t - u}{m_0^2 - s} + \frac{t - s}{m_0^2 - u} \right). \tag{6.10}$$

Theorem 6.2. *For sufficiently small r, uniformly on (6.9),*

$$|\mathcal{M}_{\mathrm{YM}} - \mathcal{M}_0| \le C_{\delta,\theta_H,\kappa_0} e^2 \left[r^2 \frac{E^2}{m_0^2 + E^2} + (EL)^2 \right]. \tag{6.11}$$

In particular, the error is $O(e^2 r^2)$ throughout this domain, independently of b/a.

Proof. Proposition 3.1 gives $m_L^2 = m_0^2[1 + O(r^2)]$; for small r, the pole separation in (6.9) therefore holds with m_L and $\delta/2$. Proposition 5.3 gives $g_*^2 j_L^2 = e^2[1 + O(r^2)]$. Replacing the light mass and current in (6.6) by their leading values costs at most

$$C_\delta e^2 r^2 \frac{E^2}{m_0^2 + E^2}.$$

The mass replacement has an additional harmless factor $m_0^2/(m_0^2 + E^2) \le 1$. The three light scalar-partner Hessians are $O(r^2)$, so their recombined contribution is bounded by $C_\delta e^2 r^4 E^2/(m_0^2 + E^2)$.

Both other low masses are at least $M_0^2[1 - Cr^2]$. The heavy-energy margin bounds their denominators below by a fixed multiple of M_0^2. Their currents and moment Hessians are $O(r^2)$, so together they contribute at most

$$Ce^2 r^4 E^2/M_0^2 \le Ce^2 r^2 (EL)^2,$$

since r^2 and a^2 are comparable. All remaining eigenvalues have the fixed gap γ. The Parseval bounds estimate their complete contribution by $Ce^2(EL)^2/\gamma$. The entire window lies below half that gap for small r. These estimates prove (6.11); finally $(EL)^2 \le Ca^2 \le Cr^2$. □

The estimate is absolute; $\mathcal{M}_0$ may vanish, so no relative amplitude bound is asserted. The permitted domain need not cross the light pole for every δ and every allowed ratio. To exhibit energies on both sides, take $0 < \delta < 1/3$ and

$$\frac{b^2}{a^2} < \theta_H \frac{1 - \delta}{1 + \delta}. \tag{6.12}$$

At physical fixed angle $t = u = -s/2$, $E^2 = s$, the above-pole interval $(1 + \delta)m_0^2/(1 - \delta) \le s \le \theta_H M_0^2$ is nonempty, as is a below-pole interval. As $b/a \to 0$, the former contains $m_0 \ll E \ll M_0$, the intermediate U(1) regime. The theorem does not estimate the resonance itself. Its vector exchange coefficient is $e^2[1 + O(r^2)]$; the same-mass scalar-partner poles have numerator coefficients $O(e^2 r^4 m_L^2)$.

6.4 Uniform curvature of the selected plane

There is also a geometric consequence of the same vertices. In real unitary L^2 normalization the quotient curvature is

$$K_{\mathbb{R}} = \sum_{n>0} \frac{1}{\mu_n} \left[3j_n^2 + \sum_{\alpha=1}^{3} \left\{ b_{n\alpha}(\mathsf{u},\mathsf{u}) b_{n\alpha}(\mathsf{v},\mathsf{v}) - b_{n\alpha}(\mathsf{u},\mathsf{v})^2 \right\} \right]. \tag{6.13}$$

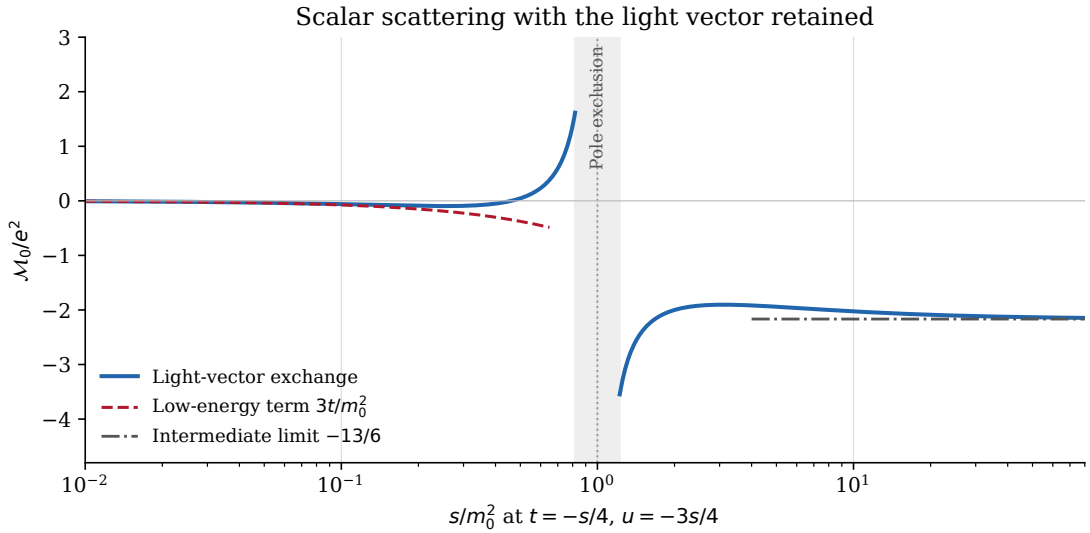


Figure 1. The explicit amplitude (6.10) at fixed angle, with $b/a = 0.1$, $\theta_H = 0.8$ and $\delta = 0.1$. The shaded interval is excluded by the light-pole separation. The low-energy curvature term and the intermediate limit describe different parts of the allowed window. These are analytic finite-model curves; the actual amplitude is compared with them by Theorem 6.2, whose error constant is not numerically evaluated.

Indeed twice differentiating the moment equations gives second fundamental form component $-b_{n\alpha}(w,z)/\sqrt{\mu_n}$ in the normal direction $n_{n\alpha}$. The Gauss equation gives the terms involving b. Differentiating the mechanical gauge connection gives O'Neill's contribution $3\sum_n j_n^2/\mu_n$ [24–26].

This calculation applies at each fixed $b > 0$ in a Sobolev slice of the stable determinant-fixed E block. Remove its ineffective constant center. The gauge and moment-normal operators then have bounded elliptic inverses at that point, defining the projections, second fundamental form and L^2 Levi-Civita connection. Neutrality removes the central cokernel; all other orthogonal-adjoint blocks have zero vertices on this plane. The complete sums converge by the fixed complementary gap and (5.12). Thus the formula does not require a lower bound uniform in b/a on the size of that local slice.

Proposition 6.3. *The sectional curvature of the actual harmonic bundle-moduli metric on the transported plane* (5.2)*, with positive sign on a sphere, satisfies*

$$K_{\mathbb{R}} = \frac{3}{4b^2}[1 + O(r^2)], \qquad K_{g^{\mathrm{KS}}} = \frac{3}{2b^2}[1 + O(r^2)], \tag{6.14}$$

uniformly on (5.1)*.*

Proof. The light vector term in (6.13) is $3/(4b^2)[1 + O(r^2)]$. The light normal terms are $O(r^4/b^2)$, the other low terms are $O(r^4/a^2)$, and the complementary sum is $O(1)$. Relative

to $1/b^2$, each error is $O(r^2)$ or smaller. The real unitary metric is twice the Dolbeault metric, so $K_{g^{\mathrm{KS}}} = 2K_{\mathbb{R}}$. □

This result concerns the specified plane. It asserts neither bounded derivatives of the full quotient metric nor uniform curvature control on every plane. Appendix B gives a complementary regular-patch derivative argument and an explicit singular horizontal projector.

7 The leading physical curvature coefficient

7.1 Partial reduction on a regular patch

Fix a relatively compact generic patch with $a, b > 0$, and allow the spectator bundle and K3 geometry to vary in small patches. Write $z = (z_E, z_B)$ for bundle coordinates, y for geometric hypermultiplet coordinates, and χ^I for locally unwrapped periodic two-form coordinates. The tensor multiplet and gravity remain explicit. At energies below the smallest Higgs mass we eliminate the three massive vector multiplets as well as the gapped KK modes. We retain all unbroken gauge and charged matter fields. The reduced scalar kinetic metric is a smooth gauged target near this vacuum; it is not the possibly singular vacuum quotient by the unbroken group.

The nondegeneracy needed for this partial reduction can be checked in the local model. Let $D : \mathfrak{su}(2) \to T_x V$ be the broken orbit map. At moment level zero its quaternionic images obey

$$\langle I_\alpha D\xi, I_\beta D\zeta\rangle = \delta_{\alpha\beta} Q_x(\xi, \zeta). \tag{7.1}$$

The symplectic pairings of orbit vectors vanish by moment-map equivariance. Since D is injective, the three moment equations have an invertible derivative on these nine real normal directions. Their norm-square potential has positive Hessian there. The three orbit directions are Goldstone directions; a local gauge slice removes them. This accounts for the twelve real scalar directions in the three massive vector multiplets.

At the actual ASD background the same construction has an elliptic description. If H is its full residual compact gauge group and $\mathfrak{h}$ its parallel algebra, the hyperkähler Weitzenböck formula identifies the ASD cokernel with $\mathfrak{h} \otimes \mathbb{R}^3$. On an H-invariant Coulomb slice the complement of this cokernel has a bounded elliptic right inverse; its finite kernel is the full retained ASD deformation space. For an off-shell effective action solve the heavy Euler–Lagrange equations on this complement. Their leading linearization is the positive ASD Hessian. The implicit-function construction gives a smooth H-equivariant solution in all retained variables, with the unbroken moment maps left as a potential. It agrees with the projected ASD construction on neutral ASD vacua; these equations need not agree away from that locus. The metric, two-form and fermionic complementary operators are those specified in section 3. All inverse bounds here belong to the fixed regular patch, and can deteriorate when its Higgs gap vanishes.

The perturbative expansion below means the finite jet of this tree action, order by order in η and external derivatives. The coefficient functions and the parameter derivatives actually

used are smooth on the fixed patch. The notation $O_{\text{formal}}(\eta^j)$ specifies the absence of lower coefficients; it does not bound an unconstructed infinite remainder at finite η.

7.2 Mixed constraints and the vertical kinetic coefficient

For a pure bundle velocity at fixed leading geometry, use a parameter gauge connection Λ_i and set

$$a_i = \partial_i \mathcal{A} - d_{\mathcal{A}} \Lambda_i, \qquad d^*_{\mathcal{A}} a_i = 0, \qquad d^+_{\mathcal{A}} a_i = 0. \tag{7.2}$$

The external-index gauge equation is precisely the first condition; $\mathcal{A}_\mu = \Lambda_i \partial_\mu z^i$ gives $F_{\mu m} = a_{i,m} \partial_\mu z^i$. For stable determinant-fixed E, B, the adjoint Laplacian is invertible off its retained central kernel. Smooth slice dependence and elliptic differentiation control Λ_i and any fixed finite number of derivatives on the patch. K3 geometric variations lift to nearby ASD connections: the trace-free stable ASD cokernel is zero, since its harmonic self-dual adjoint forms have parallel coefficients. The neutral central equations are compatible with the fixed trivial determinants.

Put $f_i = \partial_i f$ and $(j_i)_m = B_F(F_{mn}, a_i{}^n)$. For the convention $G_{\mu m} = \eta v_{i,m} \partial_\mu z^i$ the mixed Einstein equation to the required order is

$$\Delta_1 v_i = df_i + j_i, \qquad d^* v_i = 0. \tag{7.3}$$

Its coefficients follow directly from (4.1): the conformal internal metric contributes $-3\eta\, df_i/2$, the dilaton contributes $+\eta\, df_i$, the mixed metric contributes $\eta d^* dv_i/2$, and the gauge stress contributes $-\eta j_i/2$. The H stress starts at η^2; the mixed R^2 source for a pure bundle velocity does also, because the leading tangent connection is independent of z.

The longitudinal equation is essential. Differentiating (4.5), using the Yang–Mills equation and trace invariance, gives

$$\Delta_0 f_i = \tfrac{1}{2} B_F(F_{mn}, (d_{\mathcal{A}} a_i)^{mn}) = B_F(F_{mn}, D^m a_i{}^n) = -d^* j_i. \tag{7.4}$$

Thus $df_i + j_i$ is co-closed. Since $H^1(S, \mathbb{R}) = 0$, (7.3) has the controlled solution $v_i = \Delta_1^{-1}(df_i + j_i)$.

For the two-form constraint, write $H_S = \eta h_1$ and $H_{\mu mn} = \eta (\mathcal{H}_i)_{mn} \partial_\mu z^i$. The mixed Bianchi identity and the external-index two-form equation give

$$d\mathcal{H}_i = \partial_i h_1 + B_F(a_i \wedge F) =: S_i, \qquad d^* \mathcal{H}_i = 0. \tag{7.5}$$

The source is closed: the derivative of the internal Bianchi identity is $d\partial_i h_1 = -B_F(d_{\mathcal{A}} a_i \wedge F)$, and the second term cancels it using $d_{\mathcal{A}} F = 0$. Since $H^3(S, \mathbb{R}) = 0$, with G_3 the ordinary Green operator,

$$\mathcal{H}_i = d^* G_3 S_i + h_i, \tag{7.6}$$

where h_i is harmonic. These harmonic components describe the retained periodic two-form coordinates and their local connection; they are not inverted. Their parameter-space curvature is the harmonic projection of the universal Chern–Weil form, including the parameter

legs of the universal gauge connection. The universal Bianchi identity makes it closed. On a contractible patch a smooth connection potential is obtained by the Poincaré homotopy operator. This accounts for parameter integrability; setting every h_i to zero independently would not do so. All computed coefficients and their finite parameter derivatives are bounded on the patch.

The action now determines the leading bundle tensor, not merely its expected scale. The mixed Yang–Mills term is order η and gives the L^2 metric of (7.2). The leading metric and dilaton are independent of z; their bundle dependence and metric compensator start at order η, so their pure-bundle kinetic contribution starts at η^2. The two-form constraint has the same order, with its periodic-coordinate connection of order one. The leading tangent connection is independent of z; its variation through the corrected metric and torsion starts at η. The curvature-squared and Lorentz Chern–Simons terms therefore do not add an order-η pure-bundle tensor. They are not omitted from base and higher-order coefficients.

If $a_i = \beta_i - \beta_i^*$ with β_i its $(0,1)$ part, the real one-form norm is twice the Dolbeault norm. Restoring the normalization (3.1) gives

$$S_{\text{scalar}} = -\frac{1}{2}\int C_B g^{\text{KS}}_{ij}\partial_\mu z^i_E \partial^\mu z^j_E + \cdots, \qquad C_B = \frac{2L^2}{g_{10}^2}. \tag{7.7}$$

This is a block of the full kinetic tensor, not yet its intrinsic or ambient curvature.

7.3 The other neutral fields and ambient curvature

After a common six-dimensional normalization, the neutral metric has the following finite-order structure:

$$\begin{aligned} G_\eta &= g_{AB}(y)dy^A dy^B + \eta h_{ij}(y,z)(dz^i + C^i_A dy^A)(dz^j + C^j_B dy^B) \\ &\quad + \eta^2 k_{IJ}(y)(d\chi^I + V^I)(d\chi^J + V^J) + E_\eta, \end{aligned} \tag{7.8}$$

$$h(y,z) = V_0^{-1}\big(g^{\text{KS}}_{E,y}(z_E) \oplus g^{\text{KS}}_{B,y}(z_B)\big). \tag{7.9}$$

The leading blocks g, h, k are positive and smooth. The connection V has both dy and dz components. Perturbative two-form shift symmetry makes the coefficient functions independent of χ. With two parameter derivatives, the formal remainder orders are

$$\begin{gathered} E_{yy} = O_{\text{formal}}(\eta), \quad E_{yz}, E_{zz} = O_{\text{formal}}(\eta^2), \\ E_{y\chi}, E_{z\chi}, E_{\chi\chi} = O_{\text{formal}}(\eta^3). \end{gathered} \tag{7.10}$$

The leading y, z cross term is included in C. The leading two-form Hodge metric depends only on y; its first correction is one order higher. These orders follow from the constraints and action count above and agree with the geometry–bundle–two-form hierarchy of [5]. The Lorentz Chern–Simons and curvature-squared contributions identified there are included at their required orders.

We spell out the curvature consequence because the small axion block cannot be treated as a bounded spectator. In the coframe of (7.8) the scales are $(1, \eta, \eta^2)$. In coordinates the inverse orders are

$$G^{yy} = O(1), \quad G^{yz} = O(1), \quad G^{zz} = \eta^{-1}h^{-1} + O(1),$$
$$G^{z\chi} = O(\eta^{-1}), \quad G^{y\chi} = O(1), \quad G^{\chi\chi} = O(\eta^{-2}).$$

For lower-index Christoffel symbols with two bundle arguments,

$$\Gamma_{z,zz} = \eta\Gamma(h)_{z,zz} + O(\eta^2), \quad \Gamma_{y,zz} = O(\eta), \quad \Gamma_{\chi,zz} = O(\eta^2).$$

The last estimate uses shift independence. In the coordinate curvature formula the second-derivative terms and the zz Christoffel contraction give $\eta R(h)$; all other contractions have order η^2 or higher, including the one with $G^{\chi\chi}$. Consequently

$$R(G_\eta)_{EEEE} = \frac{\eta}{V_0} R(g_E^{\mathrm{KS}}) + O_{\mathrm{formal}}(\eta^2). \tag{7.11}$$

The leading product in (7.9) is essential for the pure-E coefficient. The spectator-bundle moduli have not been declared constant. A nonsingular tensor-field block can also be retained; its four-E contributions start one order later by the same calculation. In product-frame units the leading sectional curvature is therefore $C_B^{-1} K_{g^{\mathrm{KS}}}$.

It remains to compare neutral curvature with the ambient scalar target including charged fields. Every additional hypermultiplet in table 1 is charged under the residual compact group. The neutral fields are exactly the E, B moduli and the geometric/two-form fields. The equivariant partial reduction constructed above gives a smooth metric on which the full residual group acts by isometries. Its fixed component is totally geodesic: a geodesic with fixed initial point and fixed tangent remains fixed by uniqueness. Thus its four-neutral curvature is the ambient scalar curvature. The residual global gauge action, including shared centers, is retained; no further finite or continuous quotient is taken. This argument does not say that the E-only slice is totally geodesic among the neutral fields; their effects have instead been included in (7.8)–(7.11).

7.4 The interaction and its order of limits

Combining (7.7), (7.11) and Proposition 6.3 gives

$$\mathcal{K}_{\mathrm{phys}}^{\mathrm{lead}} = \frac{3g_{10}^2}{4L^2b^2}[1 + O(r^2)], \qquad m_L^2 \mathcal{K}_{\mathrm{phys}}^{\mathrm{lead}} = 3e^2[1 + O(r^2)], \tag{7.12}$$

uniformly on (5.1). In physical scalar normal coordinates with the convention assigning positive curvature to a sphere, $G_{ij} = \delta_{ij} - \frac{1}{3}R_{ikjl}\phi^k\phi^l + \cdots$. The two-polarization part of the mostly-plus Lorentzian action $-\frac{1}{2}G_{ij}\partial\phi^i\partial\phi^j$ is

$$\mathcal{L}_4 = \frac{\mathcal{K}_{\mathrm{phys}}^{\mathrm{lead}}}{6}\left[(\phi^v)^2(\partial\phi^u)^2 + (\phi^u)^2(\partial\phi^v)^2 - 2\phi^u\phi^v\partial\phi^u\cdot\partial\phi^v\right] \tag{7.13}$$

at the calculated large-volume order. It is a nonzero curvature-controlled scalar coefficient in the covariant tree amplitude dictionary [6]. Neutral exchange contributions that appear in other coordinates assemble into this coefficient; universal graviton-exchange poles are separate.

The statement first computes the leading tree large-volume coefficient at each fixed regular point. The resulting coefficient obeys the geometric relative estimate $O(a^2+b^2)$ uniformly as $b/a \to 0$, by the spectral curvature formula; no uniform parameter derivative estimate for the whole neutral kinetic tensor is needed. The derivative expansion requires $E_{\mathrm{probe}} \ll b/L$, with its next parameter $E^2_{\mathrm{probe}}/m^2_{\mathrm{min}}$, and perturbative interactions in canonical normalization. Taking the probe energy sufficiently small and the ten-dimensional coupling weak leaves a nonempty regime. Keeping a six-dimensional dilaton fixed as volume grows need not keep that ten-dimensional coupling small, so these limits are specified separately. We have not bounded higher physical coefficients near enhancement or proved a uniform simultaneous finite-η curvature remainder. At energies comparable to or above b/L, Theorem 6.2 retains the light multiplet explicitly and replaces the quotient derivative expansion.

8 Partial results at the next heterotic tree order

The amplitude in Theorem 6.2 includes the complete Yang–Mills tower. A further coefficient is required to include the other heterotic interactions. We identify its perturbative order, bound the quadratic potential mixing with K3 shape deformations, and calculate the universal massless spin-two contribution. The remaining coefficient is not evaluated here.

8.1 Order counting and the required terms

Remove the common gravitational prefactor and write the action in reference internal units as

$$S[Y,\mathcal{A}] = S_0[Y] + \eta S_1[Y,\mathcal{A}] + \eta^2 S_2[Y,\mathcal{A}] + O_{\mathrm{formal}}(\eta^3), \tag{8.1}$$

where Y denotes the metric, dilaton and two-form. The term S_1 contains both Yang–Mills and tangent curvature squared. In the field scheme of (4.1), S_2 includes the terms generated by the recursive Chern–Simons definition of H and the composite Hull connection. The absence of a new independent bosonic invariant at this order does not set S_2 to zero [21, section 3.1][10, 11, section 2.1].

Each canonical gauge or connection-scalar external leg removes a factor $\eta^{1/2}$ in these units. A bare term $\eta^n S_n$ with four such legs has order η^{n-2}. The leading Yang–Mills amplitude has order η^{-1}; its first relative-η correction requires the complete bare action through η^2. The same count follows for exchange diagrams after canonical normalization of internal fields. Independent bare η^3 terms affect a later coefficient at fixed deformation and momentum. This order count gives no estimate uniform in a vanishing Higgs mass or near a pole.

One can organize the calculation by the stationary action. Let Y_0 be the leading product background, K the gauge-fixed Hessian of S_0, and $J[\mathcal{A}] = \delta_Y S_1[Y_0,\mathcal{A}]$. With constraints, zero

modes and propagator boundary conditions specified, eliminating the neutral fluctuations to this order would give

$$W[\mathcal{A}] = S_2[Y_0, \mathcal{A}] - \tfrac{1}{2}\langle J[\mathcal{A}], K_F^{-1} J[\mathcal{A}]\rangle. \tag{8.2}$$

This follows by inserting the stationary displacement $-\eta K_F^{-1} J$ into the quadratic leading action and its source. The massless poles in K_F^{-1} are retained; the expression is not a local elimination of massless fields. At zero external momentum, the compatible complementary inverse is used, with background zero modes fixed by the stated normalization.

Formula (8.2) is an organization of the missing calculation, not a computed amplitude. If J_d is the degree-d fluctuation part, its quartic term involves $J_0 K_F^{-1} J_4$, $J_1 K_F^{-1} J_3$ and $\frac{1}{2} J_2 K_F^{-1} J_2$. The first two include the corrected background and linear mixing. One must also insert the cubic part of W into Yang–Mills exchange diagrams, insert its quadratic part into internal propagators, and correct the external modes and residues. Adding only neutral exchange between two scalar-pair sources omits these terms. The scalar constraints in section 7 and the vector pencil in section 4 supply parts of this reduction, but neither determines the complete cubic and quartic vertices.

8.2 Quadratic mixing with K3 metric deformations

One part of the coupled reduction admits a relative estimate from the same harmonic energy bound. Consider a volume-preserving infinitesimal hyperkähler metric deformation in harmonic gauge, modulo diffeomorphisms and rotations of the parallel self-dual triple. Its period variations are harmonic ASD two-forms $\beta_\alpha = (\partial_\tau \omega_\alpha)^-$; this is the usual $\mathbb{R}^3 \otimes \mathcal{H}_2^-(S)$ description of the 57-dimensional space of K3 shape tangents [27]. Normalize $\sum_\alpha \|\beta_\alpha\|^2 \le 1$.

At fixed connection, put $f_\alpha = \partial_\tau \mu_\alpha$ and $c_{n\alpha} = \langle p_n, f_\alpha\rangle$. Up to the fixed component signs in (6.2), the source is $f_\alpha = *_0(F \wedge \beta_\alpha)$. The variation of the outer Hodge star contributes zero because $F \wedge \omega_\alpha = 0$ at the ASD background, and the self-dual part of the period variation wedges to zero with F. Since β_α is harmonic and ASD,

$$|c_{n\alpha}| = \left| \int_S B_F(p_n, F) \wedge \beta_\alpha \right| = \left| \langle \mathsf{H}_2 B_F(p_n, F), \beta_\alpha \rangle \right|. \tag{8.3}$$

Lemma 8.1. *For each low positive mode,*

$$\sum_\alpha |c_{n\alpha}|^2 \le C r^2 \mu_n. \tag{8.4}$$

The mixed Hessian of $\frac{1}{2}\sum_\alpha \|\mu_\alpha\|^2$ between the unit scalar normal $n_{n\alpha}$ and this metric tangent is $\sqrt{\mu_n} c_{n\alpha}$, hence is $O(r\mu_n)$. The normal gauge-field adjustment to the linearized ASD equation is

$$\dot{\mathcal{A}}_\perp = -\sum_{n>0,\alpha} \frac{c_{n\alpha}}{\sqrt{\mu_n}} n_{n\alpha}. \tag{8.5}$$

Its low component is $O(r)$ and its complementary component is $O(1)$ in every fixed spatial Sobolev norm, uniformly in b/a.

Proof. Equations (8.3) and (4.18) give (8.4) by Cauchy–Schwarz. They also show that the source has zero projection on every parallel zero-form. At zero moment level the potential Hessian is the pairing of first moment derivatives. Equation (6.5) gives the displayed mixed entry and verifies that (8.5) solves $D\mu_\alpha(\dot{\mathcal{A}}_\perp) + f_\alpha = 0$.

For the full adjoint ASD complex the moment-map cokernel consists of three copies of the parallel algebra; the preceding vanishing removes each compatibility obstruction. The full lift includes the spectator B connection's curvature source and its gapped response. In a trace-free determinant-fixed stable E block the obstruction space is zero; restricting the lift to that block means restricting the source as well.

Orthonormality of the normals gives

$$\|\dot{\mathcal{A}}_{\perp,\mathrm{low}}\|^2 = \sum_{n\,\mathrm{low},\alpha} \frac{|c_{n\alpha}|^2}{\mu_n} \le Cr^2.$$

Each unit low normal solves the uniformly elliptic ASD one-form square with bounded eigenvalue. Its coefficients are uniformly smooth in the fixed unitary presentation, so elliptic regularity bounds the normal in every fixed H^s norm. The finite low sum therefore has the same $O(r)$ bound in H^s; dividing a spatial estimate for Dp_n by a small $\sqrt{\mu_n}$ is unnecessary. For the complementary part, Parseval and the fixed positive gap give a uniform L^2 bound. The curvature and normalized harmonic β_α have uniformly bounded spatial derivatives, so the source does also. The complementary ASD equation and elliptic regularity then give its fixed-H^s bounds. □

In the common normalization of (8.1), a canonical gauge normal carries $\eta^{-1/2}$ and a geometry mode has leading order-one normalization. Thus this potential block has a low mixed entry $O(\sqrt{\eta}\, r\mu_n)$ and a static canonical normal admixture $O(\sqrt{\eta}\, r)$. This controls one quadratic contribution. It does not bound parameter derivatives of (8.5), the full corrected kinetic form, or a two-external-scalar neutral source at nonzero momentum. Those are still needed for the complete four-point coefficient.

8.3 The massless graviton and the angular domain

Use the six-dimensional Einstein frame normalized to agree with the product-frame external metric at the fixed leading vacuum. Define

$$S_{\mathrm{EH}} = \frac{1}{2\kappa_6^2}\int \sqrt{-g}\, R, \qquad g_{\mu\nu} = \eta_{\mu\nu} + 2\kappa_6 h_{\mu\nu}. \tag{8.6}$$

The curvature convention in this Lorentzian action gives the usual healthy graviton propagator. For canonical orthogonal massless scalars, the minimally coupled graviton vertex is diagonal in species. Hence $\phi\chi \to \phi\chi$ has only t-channel graviton exchange at this order. It is

$$\mathcal{M}_{\mathrm{grav}} = \kappa_6^2 \frac{su}{t}. \tag{8.7}$$

For completeness, the conserved scalar stress at a vertex is $T_{\mu\nu}(p,p') = p_\mu p'_\nu + p_\nu p'_\mu - \eta_{\mu\nu} p\cdot p'$. The conserved-source propagator numerator in D dimensions is

$$P_{\mu\nu,\rho\sigma} = \tfrac{1}{2}(\eta_{\mu\rho}\eta_{\nu\sigma} + \eta_{\mu\sigma}\eta_{\nu\rho}) - \frac{\eta_{\mu\nu}\eta_{\rho\sigma}}{D-2}.$$

On-shell contractions give $T_1 P T_2 = (s^2+u^2-t^2)/2 = -su$. In mostly-plus signature the exchanged momentum has $q^2 = -t$, and the two vertices and propagator give $\mathcal{M} = \kappa_6^2 T_1 P T_2/q^2 = -\kappa_6^2 T_1 P T_2/t$, proving (8.7) also in $D=6$. Conservation removes longitudinal gauge terms.

Restoring the common normalization of (4.1), the gauge term is

$$-\frac{\alpha' e^{-2\Phi_0}}{8\kappa_{10}^2} B_F(F_{MN}, F^{MN}).$$

Comparison with (3.1) and integration over the leading volume $V_0 L^4$ give

$$g_{10}^2 = \frac{2\kappa_{10}^2 e^{2\Phi_0}}{\alpha'}, \qquad \kappa_6^2 = \frac{\kappa_{10}^2 e^{2\Phi_0}}{V_0 L^4} = \frac{3}{4}\alpha' e^2. \tag{8.8}$$

These are the leading background normalizations needed for this coefficient. If $|t| \geq \delta E^2$, then

$$|\mathcal{M}_{\text{grav}}| \leq \frac{3}{4\delta} e^2 \eta (EL)^2. \tag{8.9}$$

At $E \asymp b/L$ this is $O(e^2\eta b^2)$, uniformly in b/a. It is an absolute bound relative to the natural amplitude scale e^2, not a relative bound against a Yang–Mills amplitude that may vanish.

The massless pole requires a change to the domain of any full-amplitude comparison. For sufficiently small nonzero s, the domain (6.9) allows $t \to 0$ with $u \to -s$. The graviton term then has the nonzero residue $-\kappa_6^2 s^2$. Thus the pure-Yang–Mills comparison cannot be extended to the unsubtracted full amplitude on precisely the same forward-allowed domain by appending a bounded error. Angular separation from massless poles, or a specified gauge-invariant pole subtraction, is necessary. This restriction does not by itself control the other terms of (8.2).

The unresolved part is the complete constrained neutral-field source, its recursively induced contacts, the corrected scalar external states, and their combined exchanges and residues. Uniform estimates would also have to control small gauge inverses and mass insertions in this sum. They do not follow from the fixed complementary gaps alone. Even a complete calculation of this coefficient would remain distinct from a bound on the full finite-α' remainder.

9 Discussion

The amplitude comparison resolves the effect of separated Higgs scales within the leading Yang–Mills compactification on this actual K3 bundle family. At energies well below b/L, its expansion gives the selected quotient-curvature interaction. At energies comparable to or

above that scale, the light vector multiplet remains explicit. The full spectral formula includes its scalar partners and all complementary internal gauge modes, with uniform absolute error below the two heavier Higgs scales and away from poles. No fixed positive lower bound on b/a is needed.

The estimates use values of actual harmonic representatives and their currents. Their reference harmonic parts have quadratic accuracy, and orthogonality removes the potentially larger mixed terms. This explains how a selected interaction and curvature can be controlled even though derivatives of the full horizontal projector diverge. The relative vector correction has a related feature: its harmonic mass form is bounded by r^2 times the gauge energy of the same profile, so a small Higgs mass creates no inverse-mass loss in the defined first-order pencil.

The complete next heterotic coefficient remains a separate calculation. Its perturbative order is fixed by the known recursive action. The quadratic potential mixing with K3 shapes and its linear normal lift obey uniform estimates, while the graviton contribution requires separation from a massless pole. The unfinished terms include corrected scalar states, neutral-field sources, induced contacts and mixed exchanges. Establishing their combined behavior near the intermediate U(1) stratum would determine whether the leading amplitude comparison extends uniformly to that additional order, or requires a restriction on the simultaneous limit. It would not alone establish a finite-string-scale remainder. Quantum corrections, resonance widths, nonzero-FI wall crossing and bubbling require further analysis.

A The vector correction on a regular ray

The estimate of Proposition 4.1 admits an explicit leading coefficient when $a = \varepsilon$, $b = \kappa\varepsilon$ with fixed $0 < \kappa < 1$. Use the fixed Coulomb presentation of (4.14) and write

$$\alpha_\varepsilon = \varepsilon u + O(\varepsilon^2), \qquad C = [u, \cdot], \quad G = (Q_0 D_0^* D_0 Q_0)^{-1}, \quad \Lambda = \Pi_0 C^* C \Pi_0.$$

Here u is the first real harmonic deformation; all remainders hold in each fixed spatial regularity. The fixed inverse G maps $Q_0 L^2$ to $Q_0 W^{2,2}$, with its higher elliptic versions. No second derivative of the finite balancing correction is used.

The low spectral space is a graph over $K_0 = \ker D_0$. Let $X_\varepsilon : K_0 \to Q_0 L^2$ be its graph map and set

$$R = -G Q_0 C^* C \Pi_0, \qquad U_\varepsilon = (I + X_\varepsilon)(I + X_\varepsilon^* X_\varepsilon)^{-1/2}. \tag{A.1}$$

Then

$$X_\varepsilon = \varepsilon^2 R + O(\varepsilon^3), \qquad U_\varepsilon = I + \varepsilon^2 R + O(\varepsilon^3). \tag{A.2}$$

Indeed (4.16) gives $X_\varepsilon = O(\varepsilon^2)$. Writing the blocks of $D_\varepsilon^* D_\varepsilon$ as A_{00}, B, L_Q, Coulomb orthogonality gives $B = \varepsilon^2 Q_0 C^* C \Pi_0 + O(\varepsilon^3)$ and $A_{00} = \varepsilon^2 \Lambda + O(\varepsilon^3)$. The graph equation $L_Q X + B = X(A_{00} + B^* X)$, with $L_Q = L_{Q,0} + O(\varepsilon)$ between fixed elliptic spaces, proves

(A.2). Elliptic regularity upgrades the finite-dimensional graph estimate to every required spatial norm. The isometric normalization changes the parallel component only at order ε^4.

Let f_0 be the zero-mean solution of (4.5) at the enhancement point, and let $h = \Delta_0^{-1} f_0$ have zero mean, with $\Delta_0 = -\nabla^2$. Define

$$Hk = \mathsf{H}_2\big(B_F(Rk, F_0) + B_F(k, u \wedge u)\big), \tag{A.3}$$

$$\langle k, Wl\rangle = \int_S h\, B_F([u,k],[u,l])\, dV_0. \tag{A.4}$$

Internal one-form indices in the second equation are contracted with g_0. The harmonic transgression (4.17) and (A.2) give

$$U_\varepsilon^* M_\varepsilon U_\varepsilon = \varepsilon^4 H^* H + O(\varepsilon^5). \tag{A.5}$$

The terms linear in a second connection coefficient are exact under harmonic projection, which is why no such coefficient appears in (A.3).

For the kinetic term, $\Pi_0 f_\varepsilon \Pi_0 = 0$ exactly. Since $f_\varepsilon = f_0 + O(\varepsilon)$,

$$U_\varepsilon^* f_\varepsilon U_\varepsilon = \varepsilon^2(\Pi_0 f_0 R + R^* f_0 \Pi_0) + O(\varepsilon^3) = -2\varepsilon^2 W + O(\varepsilon^3).$$

For the last equality use $G(f_0 k) = hk$ for parallel k: $D_0^* D_0(hk) = (\Delta_0 h)k = f_0 k$. Self-adjointness of G then gives $\langle k, f_0 R l\rangle = -\int h\, B_F([u,k],[u,l])$.

There is a further cancellation in the energy-dependent term. Parallelness implies $[F_0, k] = 0$, so trace invariance gives $I_0^*[u,k] = 0$ pointwise. Thus

$$J_\varepsilon U_\varepsilon = I_\varepsilon^* D_\varepsilon U_\varepsilon = O(\varepsilon^2).$$

The resolvent term in Z_ε is consequently $O(\varepsilon^4)$ on the fixed interval. Combining these identities yields

$$\varepsilon^{-4} U_\varepsilon^* E_\varepsilon(\varepsilon^2 z) U_\varepsilon \longrightarrow H^* H + 2zW \tag{A.6}$$

uniformly for z in compact sets. This is an operator statement and remains valid at a repeated leading eigenvalue.

For the three simple positive branches let k_j be the unit Λ eigenvectors with $\ell_j = 8(\kappa^2, 1, 1+\kappa^2)_j/(3V_0)$, and define

$$\sigma_j = \|Hk_j\|^2 + 2\ell_j \int_S h\, |[u,k_j]|^2\, dV_0. \tag{A.7}$$

If $\mu_{\varepsilon,j}$ denotes the corresponding actual leading gauge eigenvalue, the roots of (4.12) obey

$$\lambda_{\varepsilon,j}^{[1]}(\eta) - \mu_{\varepsilon,j} = \eta\varepsilon^4 \sigma_j + O_\kappa(\eta\varepsilon^5 + \eta^2\varepsilon^6). \tag{A.8}$$

To justify the remainder, pull the normalized leading eigenvector back by U_ε. Its distance from k_j is $O_\kappa(\varepsilon)$ and its spectral separation is at least $c_\kappa \varepsilon^2$. Differentiating the simple pencil root at $\eta = 0$ gives $\lambda_j'(0) = \langle p_j, E_\varepsilon(\mu_j) p_j\rangle$. Equation (A.6) gives $\lambda_j'(0) = \varepsilon^4 \sigma_j + O_\kappa(\varepsilon^5)$. On

the low-root interval, $E = O(\varepsilon^4)$, $E_\lambda = O(\varepsilon^2)$ and $E_{\lambda\lambda} = O(\varepsilon^4)$. The separated eigenvector equation therefore gives $\partial_\eta p_j = O_\kappa(\varepsilon^2)$ and $\partial_\eta^2 \lambda_j = O_\kappa(\varepsilon^6)$. Taylor's theorem in η proves (A.8).

The Pauli family has an additional selection rule. Write $u = U \otimes \sigma_3 + \kappa V \otimes \sigma_1$, with U, V fixed unitary adjoint-valued one-forms on A. For the generators $\widetilde{k}_i = I_A \otimes i\sigma_i/\sqrt{6V_0}$,

$$H\widetilde{k}_1 = H\widetilde{k}_3 = 0. \tag{A.9}$$

Products in $u \wedge u$ have multiplicity factors in $\mathrm{span}\{I, \sigma_2\}$, so their pairings with $\widetilde{k}_1, \widetilde{k}_3$ vanish. For these two inputs, the double commutator $C^*C\widetilde{k}_i$ has multiplicity factors only in $\mathrm{span}\{\sigma_1, \sigma_3\}$. The fixed complementary projection and Green operator preserve that subspace; pairing with $F_A \otimes I$ is again zero. This proves both terms of (A.3) vanish. The smallest and middle branches thus have only the warping integral in (A.7).

The formula determines a background-dependent coefficient, not a universal nonzero number. Its value requires the reference ASD fields and normalized harmonic representatives. Zero mean does not give the warping integral a sign. At $\kappa = 1$, individual branch coefficients require the geometric splitting on the repeated leading eigenspace as well as (A.6). Finally (A.8) concerns the defined pencil: its powers of η do not bound uncomputed string corrections.

B Regular-patch derivatives and the quotient model

B.1 Preserving the quadratic rate in holomorphic coordinates

The direct spectral proof in section 6 controls the selected curvature uniformly in b/a. A complementary approach on regular angular patches upgrades (2.10) to metric derivative estimates. Its quadratic rate cannot simply be differentiated; the additional input is that both metrics are Ricci-flat Kähler in the same complex structure.

Lemma B.1. *Let g_* be a smooth Ricci-flat Kähler metric on a complex manifold U, and let $K \Subset U_1 \Subset U$. Suppose g_t are Ricci-flat Kähler for the same complex structure and $\sup_{U_1} |g_t - g_*|_{g_*} \le e_t \to 0$. Then for every fixed integer $k \ge 0$ and $0 < \alpha < 1$,*

$$\|g_t - g_*\|_{C^{k,\alpha}(K, g_*)} \le C_{K, U_1, g_*, k, \alpha} e_t \tag{B.1}$$

for all sufficiently small t. The comparison uses the original complex coordinates, without a parameter-dependent coordinate change.

Proof. Cover K by finitely many fixed holomorphic coordinate balls compactly contained in larger balls in U_1. Uniform metric equivalence and [28, Proposition 2.1] give uniform bounds for every fixed coordinate derivative on intermediate balls. These estimates alone do not preserve the rate e_t.

For a Ricci-flat Kähler metric, differentiation of $\log\det g$ and the Kähler interchange of derivatives give

$$g^{p\bar{q}} \partial_p \partial_{\bar{q}} g_{i\bar{j}} = g^{p\bar{q}} g^{r\bar{s}} (\partial_i g_{p\bar{s}})(\partial_{\bar{j}} g_{r\bar{q}}). \tag{B.2}$$

Subtract the equations for g_t and g_*, and write $h = g_t - g_*$. The inverse difference is $g_*^{-1} - g_t^{-1} = g_t^{-1} h g_*^{-1}$. Integrating the differential of the quadratic right side along $g_* + sh$, $0 \le s \le 1$, gives a homogeneous linear system

$$g_t^{p\bar{q}} \partial_p \partial_{\bar{q}} h = A_t \cdot Dh + B_t h. \tag{B.3}$$

D here includes all real first derivatives. The principal symbol is scalar times the identity on Hermitian components. Its ellipticity and all needed coefficient norms are uniform on intermediate balls by the preceding regularity estimate.

For clarity, lower-order coupling does not require a maximum principle for a system. Apply scalar interior Schauder estimates componentwise in the usual boundary-distance-weighted Hölder norms on an intermediate ball. They give

$$|h|^*_{2,\alpha} \le C\|h\|_0 + C_1\big(|h|^*_{1,\alpha} + |h|^*_{0,\alpha}\big).$$

Weighted interpolation bounds the parenthesis by $\delta |h|^*_{2,\alpha} + C_\delta \|h\|_0$. Choose $C_1\delta < 1/2$ and absorb. The weights are bounded below on a smaller ball, yielding the ordinary $C^{2,\alpha}$ bound by $\|h\|_0 \le e_t$. Differentiating (B.3) and using nested balls gives each higher order inductively, since every differentiated forcing term has the already established rate. The finite cover proves (B.1). □

Let δ_ε be cone dilation and put

$$g_\varepsilon = \varepsilon^{-2} \delta_\varepsilon^* \Psi^* g^{\mathrm{KS}}. \tag{B.4}$$

On any compact subset K of the smooth-simple annulus $\mathcal{C} \cap \{1/4 < r < 4\}$, both g_ε and $g_{\mathcal{C}}$ are Ricci-flat Kähler in the same holomorphic chart. Equation (2.10), on a slightly larger regular neighborhood, and Lemma B.1 give

$$\|g_\varepsilon - g_{\mathcal{C}}\|_{C^{k,\alpha}(K, g_{\mathcal{C}})} \le C_{K,k,\alpha} \varepsilon^2. \tag{B.5}$$

This is a standard elliptic consequence of the imported quadratic comparison, not a new general regularity theorem.

The dependence on the smooth chart can also be made explicit. Call a holomorphic embedding $\varphi : B_{\mathbb{C}^n}(0,2) \to \mathcal{C}_{\mathrm{simple}}$ admissible at scale $0 < \rho \le 1$ if its image lies in $1/4 < r < 4$ and

$$\Lambda^{-1} g_{\mathrm{Eucl}} \le \rho^{-2} \varphi^* g_{\mathcal{C}} \le \Lambda g_{\mathrm{Eucl}} \tag{B.6}$$

for a fixed Λ. Apply the same proof to the two pulled-back metrics divided by ρ^2. The reference derivatives are uniformly bounded by the same local Ricci-flat estimate, so at the chart center

$$\begin{aligned} |\nabla^k (g_\varepsilon - g_{\mathcal{C}})|_{g_{\mathcal{C}}} &\le C_k \varepsilon^2 \rho^{-k}, \\ |\operatorname{Rm}(g_\varepsilon) - \operatorname{Rm}(g_{\mathcal{C}})|_{g_{\mathcal{C}}} &\le C \varepsilon^2 \rho^{-2}. \end{aligned} \tag{B.7}$$

Use the same tensor type on both sides of the curvature difference. No lower bound for this admissible ρ in terms of b/a is asserted. Such a lower bound, not merely the existence of very small charts, would be needed to turn (B.7) into an explicit estimate uniform toward the Pauli axis.

B.2 What fails near a changing stabilizer

There is a direct diagnostic within the balanced locus. Add $X_3 = c\sigma_2$ to (2.11), with a, b, c real and $Y = 0$. On the real two-plane of components $(X_2\sigma_2, X_3\sigma_1)$, the orthogonal projection onto the orbit of $i\sigma_3$ is

$$P(b,c) = \frac{1}{b^2+c^2}\begin{pmatrix} b^2 & -bc \\ -bc & c^2 \end{pmatrix}. \tag{B.8}$$

The other orbit generators occupy mutually orthogonal component planes. At $c = 0$,

$$\partial_c P = \begin{pmatrix} 0 & -b^{-1} \\ -b^{-1} & 0 \end{pmatrix}, \qquad \partial_c^2 P = \begin{pmatrix} -2b^{-2} & 0 \\ 0 & 2b^{-2} \end{pmatrix}. \tag{B.9}$$

Their operator norms are b^{-1} and $2b^{-2}$. Thus the horizontal projection in the original balanced coordinates is not uniformly C^2 as $b \to 0$. This does not invalidate the relative C^0 tensor comparison, and alone it is not an intrinsic curvature calculation. The following calculation supplies the latter.

B.3 Curvature in additional harmonic deformation directions

At a Pauli point with $a, b > 0$, let u, v be the two Dolbeault vectors in (5.2). They are orthonormal for (2.6). Their support is orthogonal to the orbit and its three quaternionic images, so they are horizontal tangent vectors to the hyperkähler quotient.

Proposition B.2. *With curvature sign positive on a round sphere,*

$$K_{g_C}(u,v) = \frac{3}{2b^2}. \tag{B.10}$$

For $a = \varepsilon$, $b = \kappa\varepsilon$ at fixed $0 < \kappa \le 1$, transport this plane by Ψ and normalize using the actual harmonic metric. Its sectional curvature satisfies

$$K_{g^{\mathrm{KS}}} = \frac{3}{2\kappa^2\varepsilon^2} + O_\kappa(1). \tag{B.11}$$

Proof. The effective PU(2) action is free at this point and the moment map is a submersion. Its zero set N is a smooth submanifold of the flat representation. Every tuple of Hermitian X_j with $Y = 0$ lies in N. The affine plane $x + su + tv$ therefore lies in N, and its three relevant second fundamental form entries vanish. The Gauss equation gives $K_N(u,v) = 0$.

Let $D_x\xi = ([\xi, X_j], [\xi, Y_j])$ and use the bilinear orbit Gram form $Q_x(\xi,\zeta) = \langle D_x\xi, D_x\zeta\rangle$, viewed as a map from the Lie algebra to its dual. For horizontal u, v, set $\beta(\xi) = \langle \xi \cdot u, v\rangle$. Differentiating the one-form $\langle D_x\xi, \cdot\rangle$ gives $2\beta(\xi)$. Thus the curvature of the mechanical

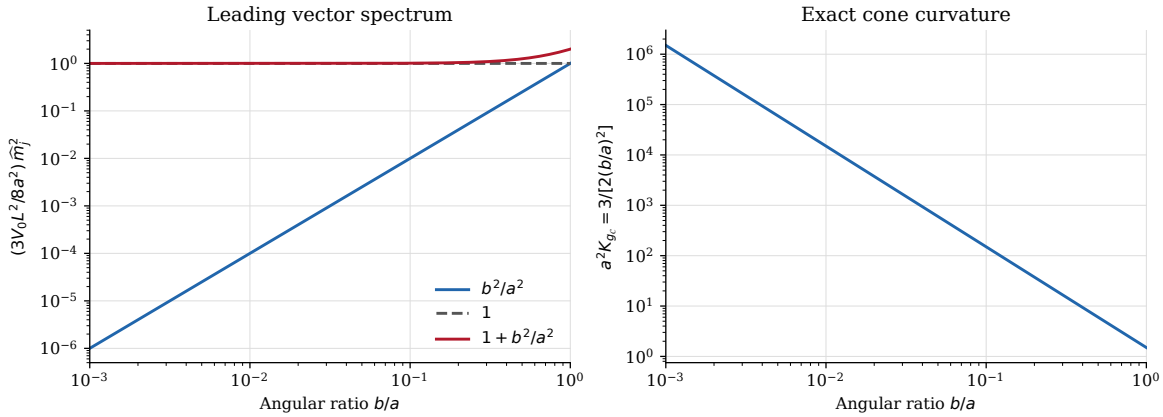


Figure 2. Exact finite-model formulas as the angular ratio decreases. Left: the three normalized leading squared vector masses. The relative geometric estimate remains uniform over the plotted ratios. Right: the cone curvature of the horizontal two-plane (5.2). Proposition 6.3 transfers this selected curvature to the actual harmonic metric with uniform relative quadratic error for $0 < b \le \kappa_0 a$, $\kappa_0 < 1$. Both panels plot analytic formulas, not numerical HE solutions.

connection is $2Q_x^{-1}\beta$, up to an irrelevant sign, and the squared norm of O'Neill's tensor is $\beta(Q_x^{-1}\beta)$. In the unnormalized basis $(i\sigma_1, i\sigma_2, i\sigma_3)$,

$$Q_x = \mathrm{diag}(8a^2, 8(a^2+b^2), 8b^2), \qquad \beta = (0, 0, -2). \tag{B.12}$$

Q_x here is the bilinear Gram matrix, not the matrix of $D_x^* D_x$ after a Hilbert–Schmidt identification of the Lie algebra with its dual. The resulting norm is $1/(2b^2)$. O'Neill's formula [24, 25] gives $K_{N/\mathrm{PU}(2)} = K_N + 3\|\mathcal{A}_{uv}\|^2$, proving (B.10).

The ray with fixed κ lies in a fixed regular angular patch; its cone radius is $\sqrt{2(1+\kappa^2)}\varepsilon$. The C^2 estimate (B.5) changes sectional curvature of the rescaled metric by $O_\kappa(\varepsilon^2)$. Scaling back gives (B.11). In particular its relative error is $O_\kappa(\varepsilon^2)$. □

Equation (B.11) supplies an independent regular-ray check of Proposition 6.3. The latter's direct vertex proof also allows $b/a \to 0$; the local derivative argument above does not provide that additional uniformity. Section 7 identifies the leading physical coefficient while retaining the other neutral fields.

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